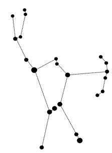

Sol: Beyond Earth

Version: 1

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The crew of the spacecraft, three astronauts and an engineer, sat calmly in their seats as they approached the space habitat. The captain, a seasoned veteran, kept a watchful eye on the controls, while the others chatted idly amongst themselves.

"Can't wait to get some real food," the engineer remarked, rubbing his stomach.

"Me too," the mechanic agreed. "I'm sick of these pre-packaged meals."

The captain chuckled. "You guys are always complaining. Just be glad we have enough to eat up here."

As they neared the space habitat, the captain switched over to the communications system and spoke in a formal, professional tone.

"This is Captain Johnson of the spaceship Odyssey. Requesting permission to dock with the habitat."

"Roger that, Captain," came the response from the control room. "Please provide your ID and docking code for verification."

The captain provided the necessary information, and after a brief pause, the control room confirmed their clearance for docking.

"Copy that, Control," the captain replied. "We are initiating docking procedures," he said, his voice calm and collected."

"Roger that, Odyssey," came the reply from the control room. "We have confirmed your approach vector. You are cleared for docking at dock 5."

"Copy that, Control. Initiating docking sequence now," the Captain responded.

As the spacecraft moved closer to the habitat, the tension in the cabin began to rise.

"Steady as she goes," the Captain muttered under his breath, his eyes fixed on the monitor in front of him.

The crew fell silent as they focused on the task at hand. They expertly maneuvered the spacecraft into position, and the captain skillfully guided it into the docking port of the habitat.

“Docking complete,” the captain announced, once the spacecraft was secured. “We’re ready for airlock transfer.”

The crew gathered their gear and made their way to the airlock, taking care to follow all safety protocols. As they waited for the airlock to cycle, they resumed their casual conversation.

“Did you guys hear about the latest mining accident on Mars?” the roboticist asked.

The engineer shook his head. “No, what happened?”

As the airlock finished cycling and the hatch opened, the captain interrupted their conversation with a stern reminder.

“Remember, guys, we’re representing our agency up here. Let’s keep our conversation professional when we’re communicating with the control room.”

The crew nodded in agreement as they made their way into the space habitat, ready to complete their mission with precision and professionalism.

The year is 2090, and the Federated Autonomy Treaty has just established the Sol Union: a federation of Earth’s major power blocs and the outer settlements, brokered under rising tensions over taxation and autonomy. Earth remains the administrative hub of this new union, with the Moon serving as a major transportation and commerce hub. Mars is undergoing terraforming, a process recognized as centuries-long, and hosts tens of thousands of settlers in pressurized habitats, research facilities, and early industrial operations. The moons of Jupiter and Saturn have been settled with research stations, mining outposts, and small colonies. Total human population beyond Earth numbers in the hundreds of thousands.

Humanity has made significant advances in propulsion, energy generation, and materials science. Fusion-powered spacecraft have cut transit times dramatically: Earth to Mars in weeks to months, depending on orbital alignment. Artificial intelligence and robotics have made it possible to build and maintain settlements in environments hostile to unprotected human life.

However, living in space is not easy. The environment is harsh, and the risk of radiation exposure, micrometeoroid impacts, and other hazards is ever-present. Resources are limited, and conflicts arise over access to water ice, helium-3, and rare minerals. The Sol Union is newly formed and already strained: Earth-centered institutions and outer settlements are locked in disputes over who pays what and who decides.

Sol: Beyond Earth takes place in this setting. Players take on the roles of astronauts, scientists, engineers, miners, and other space-faring professionals. They will face a variety of challenges, from repairing damaged spacecraft to negotiating with rival factions over access to resources. They will need to use their skills and knowledge to survive in this harsh environment, managing the competing political pressures of the solar system.

What is Sol: Beyond Earth

Sol: Beyond Earth is a roleplaying game in which you will define an imagined space bounded by rules. Actions and events occur through procedure-driven player interaction. This facilitates an emergent shared narrative.

Put plainly, players will assume the roles of characters in a fictional setting. The characters will act according to the rules of the game that will determine their success or failure. The outcome of actions will have an effect on the narrative that emerges.

A game of *Sol: Beyond Earth* takes place as a conversation between the Referee and the other players.

- The Referee sets up the situation, describes the environment and setting, moves the dialogue and actions of the Non-Playing Characters.

- The Players move in the imagined space of the game through their Playing Character, describing their actions and narrating their conversations. It is up to each player to decide whether to act out their Character or play them impersonally. Players should also ask questions of clarification to the Referee.

To play you will need:

- At least 2 players are needed; one takes the role of Referee. The ideal group is 4–5 players total, maximum 6.
- A complete set of polyhedral dice, preferably one for each player.
- Writing instruments, index cards and sheets of paper for note-taking.

Game Principles

It is important that everyone at the table feels comfortable and has the following principles in mind:

- **Fiction over mechanics.** This ruleset is intentionally streamlined and prioritizes fiction over complex mechanics. If you are primarily interested in a game with detailed mechanical systems, there are other rulesets that may be better suited to your preferences.
- **The Referee is a player.** Although the Referee has different responsibilities than the other players, they should still be considered as a fellow player and should not shoulder more responsibility for the success of the game than anyone else at the table.
- **Referee's Mediating Role.** The Referee does not compete against other players; their role is solely to arbitrate situations within the context of the game. They are responsible for maintaining consistency, ensuring fairness, and resolving any disputes or conflicts that may arise during play.
- **Promote Cooperative Play.** This game is intended to be played collaboratively, with a focus on teamwork and overcoming challenges as a group. As such, competition between players should be discouraged in favor of cooperation and mutual support. The characters should work together to confront the obstacles and threats posed by the game world, rather than engaging in individualistic or adversarial behavior.

- **Information Query.** Players should ask the Referee questions in a concise and straightforward manner. The Referee, in turn, should provide clear and detailed explanations of the context and the potential risks associated with each action taken by the characters. This helps to ensure that players make informed decisions and understand the consequences of their actions.
- **Action first.** When describing an action, players should first provide a clear and detailed description of what their character is attempting to do. Only after the action has been fully described should the Referee request a dice roll, if necessary, to determine the outcome of the action. This helps to ensure that the narrative flows smoothly and that players have a clear understanding of the sequence of events.
- **Emergent Fiction.** The game thrives on emergent fiction, where the narrative evolves naturally from player choices and interactions. Be prepared for unexpected twists and developments, and allow the story to unfold organically based on the characters' actions and decisions. Embrace the creativity and spontaneity that arise from collaborative storytelling, and don't be afraid to deviate from preconceived notions of how the plot should progress.
- **Embrace uncertainty.** Uncertainty drives the game. Embrace the unpredictability of player choices and the twists and turns the narrative may take. Avoid rigidly planning every detail and be open to adapting to the unexpected. The most memorable moments tend to come from outcomes no one planned for.

A word about "realism"

Consistency matters more than simulation. The setting, equipment, and factions follow internal logic; keep decisions within what physics and the established world make possible, and avoid choices that only work if you ignore both. Prior knowledge of the hard sci-fi genre helps but is not required; familiarity with the inspirations listed at the back deepens appreciation for some of the design choices.

Safety Tools

It is important to remember that not everyone at the table may be comfortable with the themes or content addressed by the game, or with the actions of the other players. It is therefore important to establish clear boundaries and expectations at the beginning of the game, and to be sensitive to the needs and feelings of all participants throughout the session. It may also be helpful to establish a system for communicating discomfort or taking breaks if needed.

There are various safety tools and techniques that can help create a comfortable and safe environment for everyone involved in the game, such as Lines And Veils, C.A.T.S., the X-Card, and Script Change.

It is important to familiarize yourself with these tools and techniques and to use them proactively to ensure that everyone feels safe and comfortable during the game.

Characters

Playable characters are the players' gateway to the game universe. By assuming their role, players will move around dictating their intentions and actions.

1) First choose the role you want to take on from:

DEBRIS CLEANER: Skilled in Piloting (d8) and Repair (d8). Take suit, removal tools, thrusters, and comms.

DIPLOMAT: Skilled in Negotiation (d8) and Networking (d8). Take credentials and comm device.

ENGINEER: Skilled in Repair (d8) and Science (d8). Take a toolbox and laptop (bulky).

EXPLORER: Skilled in Piloting (d8) and Science (d8). Take a spacecraft, instruments, and survival gear (bulky).

PILOT: Skilled in Navigation (d8) and Piloting (d8). Take a flight suit and helmet.

MEDIC: Skilled in Medicine (d8) and Engineering (d8). Take a medkit and surgical tools (bulky).

MARINE: Skilled in Athletics (d8) and Firearms (d8). Take a vacuum suit and rifle (bulky).

SCIENCE OFFICER: Skilled in Computers (d8) and Science (d8). Take sensor gear and a tablet (bulky).

SPACE MINER: Skilled in Engineering (d8) and Resources (d8). Take a mining suit, portable drill, and scanner (bulky).

2) Pick 3 more skills increases (from no skill->d8->d10->d12):

Athletics, CQC (Close Quarters Combat), Computers, Engineering, Firearms, G-Zero, Leadership, Linguistics, Medicine, Navigation, Negotiation, Networking, Piloting, Psychology, Repair, Resources, Robotics, Science

Details

Customize details to fit the setting. Here are some options fitting a hard sci-fi space crew:

Name

1	Alex	6	Cameron	11	Casey	16	Chris
2	Dakota	7	Dana	12	Devon	17	Drew
3	Elijah	8	Erin	13	Hayden	18	Jackie
4	Jamie	9	Jess	14	Jo	19	Jordan
5	Kelly	10	Kris	15	Morgan	20	Skyler

Surname

1	Abrams	6	Chen	11	Choi	16	Clarke
2	Dubois	7	Khan	12	Kowalski	17	Patel
3	Sato	8	Singh	13	Tran	18	Wright
4	Yoshida	9	Zhang	14	Martinez	19	Rodriguez
5	Garcia	10	Lopez	15	Gomez	20	Gonzales

Nickname

1	Ace	6	Buzz	11	Cap	16	Doc
2	Dragon	7	Eagle	12	Firefly	17	Gemini
3	Goose	8	Hawk	13	Huntress	18	Jet

1	**Ace**	**6**	**Buzz**	**11**	**Cap**	**16**	**Doc**
4	Kick	9	Lightning	14	Maverick	19	Red
5	Rocket	10	Shadow	15	Solo	20	Top Gun

Demeanor

1	**Adventurous**	**6**	**Ambitious**	**11**	**Brave**	**16**	**Brusque**
2	Cautious	7	Composed	12	Curious	17	Focused
3	Intense	8	Jovial	13	Methodical	18	Patriotic
4	Practical	9	Professional	14	Resourceful	19	Skeptical
5	Stoic	10	Witty	15	Wry	20	Zealous

Quirks

1	**Launch buff**	**6**	**Lucky socks**	**11**	**Conspiracy theorist**	**16**	**Contraband hooch**
2	Homesick	7	Compulsive journaler	12	Meticulous documenter	17	Knuckle cracker
3	Spacesick	8	Green thumb	13	Washout grudge	18	Claustrophobic
4	Diet zealot	9	Sports nut	14	Retrogamer	19	Illegal tunes
5	Pizza lover	10	Divorcing	15	Midlife crisis	20	Last mission

Gear

You start with a communicator, vacuum suit, and 2 Supply Credit (₵) to start. Most items cost 1₵ each.

TOOLS: Chem kit, demolition charges, EVA jetpack, laser cutter, mining charges, monocular, motion tracker, repair kit, rock hammer, survey drones.

WEAPONS: Pistol, rifle, shotgun (bulky), stun baton.

ARMOR: Flak jacket, combat armor (bulky; bulky helmet grants oxygen).

CYBERNETICS: Cybernetic leg (faster movement), eye (enhanced vision), arm (increased strength).

How to play

Sol: Beyond Earth is built for mission-based play. Each session centers on a specific job or objective with a clear scope. This keeps the system's risk-driven resolution working as intended. The rules are calibrated for situations where something is at stake, and a mission frame ensures something usually is. Extended low-pressure stretches work best as transitions between missions, not as the main mode of play.

The role of the Referee is to control the game universe: he describes situations, NPCs, and their reactions to the PCs' actions. He or she should be as clear as possible about the conditions of the context in which the characters are immersed: if there is a reasonable certainty that the character has access to a piece of information or knowledge, the referee should communicate it to the player, without asking for a roll. In this game, a roll is made only for situations of risk or uncertainty of the outcome of a direct action.

- If an action is impossible, the referee simply declares it so by explaining why and describing the possible alternatives to the player.
- If the cost of an action is explicit, the referee will clearly present it. The player can decide whether to take the action or try another one.
- If the action involves a risk, the referee will communicate this clearly. The player will be able to make informed decisions about the action they want to take and they will pull to determine the outcome.

In all other cases, the action is automatically successful.

Rolling

When a character attempts an action with significant risk, the player rolls a skill die to determine the outcome. The default is a d6.

- If the character has applicable skills or talents, they roll a bigger die like d8 or d10 to represent competence.
- Conversely, a character *hindered* by injuries or obstacles rolls a smaller die like d4 to show impairment.

The player can also roll bonus dice based on circumstances:

- Favorable conditions grant an extra d6 die. Things like having the high ground in combat or proper tools for a repair task.
- An ally who helps adds their own skill die to the roll, representing combined effort. The ally shares in any risk.

The player rolls their skill die plus any bonus dice, taking the single highest result.

- **1-2, disaster.** The full brunt of the risk occurs, and the Referee judges if there is any success at all. A roll this bad could mean instant death when mortality is on the line.
- **3-4, setback.** You suffer a lesser or partial consequence. If risking death, the character might survive but with a grievous injury and *hindered.*
- **5+, success.** The higher numbers, the better the outcomes.

The core mechanic balances risk versus reward, with competence and teamwork aiding success. But dire consequences are still possible, maintaining high drama and tension.

Situation

When characters act without risk, the GM rolls a die sized to how settled things are: d4 (chaotic), d6 (precarious), d8 (manageable), d10 (predictable), d12 (stable), to find how the world moves.

- 1–2 Trouble. An adverse fact surfaces, something the characters can see and respond to.
- 3–4 Signs. Trouble shows at the edge; the situation grows less stable, the next die will likely be smaller.

- 5+ Calm. Nothing new; the action lands.

The surfaced fact must be proportionate to what is established, leave something to act on, and follow from prior events, not invented to fill the silence.

Opponents

Characters played by the GM have no skill dice. Define them by their **behavior** (what they want, including when they would rather flee or bargain than fight), the **risks** they present (the 1–2 result for players in conflict with them), and any **obstacles** (visible conditions, armor, numbers, cover, that must be cleared before they can be put down). Signal both before players roll.

Conflict

In a fight, the GM rolls the situation die: d4 if ambushed or scattered, d6 if evenly matched, d8 if holding an advantage, d10 if dominating, d12 if the opposition is broken, to find what the opposition does.

- 1–2 Press. A new risk or obstacle surfaces; the fight grows harder.
- 3–4 Shift. The opposition adapts, repositions, changes tactic; the situation tilts.
- 5+ Opening. An obstacle clears, a gap appears, or the opposition is ready to break.

Roll in lulls, at the start of a fight, or whenever no skill roll applies. For extended conflicts, use Steps: state actions and risks, roll as needed, describe how the situation has changed, repeat. On an Opening with the opposition cornered, they break per their behavior, fleeing, yielding, or bargaining.

Load

Characters can carry as many possessions as reasonably fits their physical capacity and the storage space on their person. For example, a character could carry a weapon, some tools, and a backpack containing rations and other useful adventuring gear.

However, if a character loads themselves down with multiple overly bulky or heavy items, it may hinder their ability to move and act freely at times. For instance, carrying a large shield, a two-handed weapon, and a bulging sack of gear may make it difficult to squeeze through narrow spaces or balance while climbing.

In such cases, the referee will inform the player that their character's encumbrance is *hindering* their movement or dexterity in the current situation. The player can then decide to temporarily set down some of their carried items to proceed unfettered, stowing the extra gear nearby to be retrieved later. Or they may opt to continue carrying everything, accepting additional challenges or risks to their intended actions.

Advancement

After successfully completing a mission or major story milestone, each player character will improve in the following ways as a reward for their accomplishments:

- Increase one of their skill ranks by one step on the progression track (none → d8 → d10 → d12). This represents honing their competencies through experience.
- Gain a number of currency credits (₵) equal to a d6 roll. Credits can be spent to acquire new gear, weapons, cyberware, services, etc. The die roll result encourages uncertainty in the exact amount of new wealth obtained.

For example, after infiltrating an enemy base undetected and stealing valuable data, a character with stealth and hacking skills currently at d8 could increase their stealth to d10 for greater sneaking ability. They also would gain somewhere between 1-6 credits to spend on desired upgrades and purchases.

Each mission adds a skill rank and an uncertain credit payout. Characters improve through use, and the randomness in credits means players can plan upgrades but not rely on them.

Damage Control

Say how you use a piece of gear to absorb damage. It's disabled until repaired at a base.

When a character is about to take damage from an attack or hazard, the player can describe how they utilize a piece of equipment to absorb some or all of the incoming damage. This could involve actions like raising a shield to block an incoming blow, using a large tool as cover from an explosion, or angling a vehicle to have an attack hit a reinforced section.

After being used to soak damage this way, that gear becomes disabled and non-functional until it can be repaired back at a home base or facility with the proper tools and parts.

The referee determines whether the gear fully negates the attack or only reduces it. High-damage or repeated attacks may exhaust a piece of gear's capacity to absorb.

Injuries

If a character sustains a serious injury during a game session, they will need to spend some in-game time in the medical bay to recover. The player should describe their character receiving medical treatment and the referee can narrate the passage of time during the healing process.

If a character is killed, the player should introduce a new character as soon as is practical. The referee should facilitate quickly integrating the new character into the ongoing story.

When adjudicating the effects of injuries or character death, inclusion of the affected player takes priority over adhering to strict realism. The referee should aim to avoid excluding a player from participation due to their character being incapacitated. Workarounds like having the player take temporary control of a supporting character can help keep everyone involved.

The risks of combat and other dangerous activities are an important part of the game's drama and stakes. However, the ultimate goal is for everyone to have fun together. When serious consequences occur, the referee should find a way to return the affected player to active participation quickly.

Mission Control

The referee portrays non-player characters (NPCs) by describing their behaviors, motivations, and obstacles they present to the player characters. The referee should lead the group in establishing boundaries for subject matter and conduct during play.

To control pacing and safety, the referee can fast-forward through uneventful stretches, pause play for breaks or sidebar discussions, or even rewind and redo a scene if needed. The referee should invite players to request scene edits too.

The referee should present dilemmas and challenges without pre-planned solutions, allowing the players to drive the direction of the story through their choices.

To give everyone screen time, the referee moves the spotlight around, checking in with each player in turn.

When careful play has kept the dice quiet for a while, test for bad luck to restore external input: roll a die. On 1–2, trouble surfaces (a patrol changes route, a contact gets nervous, a resource runs out). On 3–4, signs of trouble ahead. Before a session of low-profile play, establish at least one ambient pressure (fuel shortage, oxygen shortage, time pressure, missing clearance, unstable weather, equipment wear, communications delay, or political scrutiny) so the bad luck check has something concrete to trigger.

When the written rules don't adequately cover a situation, the referee improvises rulings in the moment to keep the game moving. If any rulings seem unsatisfactory, the group should discuss them on a break and revise them collaboratively.

The referee portrays the world and inhabitants surrounding the player characters, sharing narrative power while guiding pacing and monitoring safety.

Playing Solo

Sol: Beyond Earth can be played alone. You take on both the protagonist and the world: making decisions as your character, then stepping back to let the setting respond through the oracles below.

The **Risk Oracle** tells you how much friction the world offers before you commit to an action. The **Question Oracle** lets the world answer questions you can't resolve alone (what an NPC does, whether the docking bay is sealed, if the cargo matches the manifest). **Sparks** give you a prompt when you need direction or momentum.

Trust the results. When an oracle contradicts what you expected, that's where the story is.

The Loop

Each time your character acts, go through these steps:

1. **Declare your action.** What are you doing, and what do you want to happen?
2. **Assess the situation.** Use the *Situation Roll* to read how much is at stake.
3. **Roll if needed.** If the Situation Roll says there's friction or volatility, make your standard roll. Otherwise, it just happens.
4. **Interpret the outcome.** Let the result shape the fiction. If you need to know how the world or an NPC responds, use the *Question Oracle*.

Question Oracle

When the outcome isn't about what your character *does* (NPC reactions, world states, things off-screen), ask a yes/no question and roll based on the odds.

Likelihood	Die
Very Unlikely	d4
Unlikely	d6
Likely	d8

Likelihood	Die
Very Likely	d10
Almost Certain	d12

Interpret the result:

- **1-2: No, and...** It doesn't happen, and things get worse.
- **3-4: Yes, but...** It happens, but with a complication.
- **5+: Yes, and...** It happens, and something else opens up.

Sparks

When you need a fresh angle or you're stuck, combine an Action and a Theme to generate a prompt.

Action

D20	Action
1	Investigate
2	Rescue
3	Sabotage
4	Pursue
5	Retrieve
6	Expose
7	Defend
8	Infiltrate
9	Negotiate
10	Escape
11	Salvage
12	Destroy
13	Protect
14	Surveil
15	Confront

D20	Action
16	Transport
17	Warn
18	Steal
19	Betray
20	Liberate

Theme

D20	Theme
1	Wreck
2	Fugitive
3	Conspiracy
4	Data
5	Agent
6	Evidence
7	Outpost
8	Syndicate
9	Contact
10	Cargo
11	Survivor
12	Rival
13	Anomaly
14	Habitat
15	Technology
16	Mutiny
17	Lockdown
18	Vacuum
19	Signal

D20	Theme
20	Claim

Interplanetary Travel

Reaching a destination is only part of interplanetary travel. Characters must also manage fuel, life support, and transit time, and handle whatever the journey throws at them: navigation errors, debris, distress signals, hostile ships, or mechanical failures.

Random encounters, hazards, and decisions along the route are as much a part of a mission as the destination itself.

Navigation Challenges

Interplanetary travel demands careful planning. Orbital dynamics, gravitational forces, and celestial obstacles all affect the route, and errors compound over millions of kilometers.

During interplanetary travel, characters may encounter various navigation challenges that can impact their journeys. These challenges may include:

1. **Precise Course Plotting:** Navigating through vast distances requires precise course plotting to ensure optimal trajectories and fuel efficiency. Characters must make navigation checks, taking into account factors such as celestial bodies, gravitational assists, and transfer windows. The outcome of these checks can determine the accuracy and efficiency of their course, potentially saving precious time and resources.

2. **Gravity Assists:** Gravity assists can be utilized to conserve fuel and increase spacecraft speed. Characters may need to identify opportunities for gravity assists by plotting trajectories that allow their spacecraft to utilize the gravitational pull of nearby planets or moons. Successfully

executing a gravity assist maneuver can significantly enhance the spacecraft's velocity and propel it toward its destination faster.

3. **Hazardous Areas:** Certain regions within the solar system may be deemed hazardous due to intense radiation, asteroid fields, or other dangerous phenomena. Characters must navigate around these areas or take precautions to mitigate the risks. Navigation checks may be required to identify safe routes or to assess the best methods of shielding the spacecraft from harmful radiation or potential collisions.
4. **Celestial Obstacles:** Interplanetary travel may bring characters face-to-face with celestial obstacles such as space debris, comets, or other unexpected objects. These obstacles can pose a threat to the spacecraft's integrity and require quick thinking and skilled maneuvering to avoid collisions or damage. Navigation checks may be necessary to react swiftly and make course adjustments to evade these obstacles.

Characters with relevant navigation skills will have a better chance of successfully overcoming these navigation challenges. Their knowledge and problem-solving abilities will aid in plotting accurate courses and identifying optimal trajectories.

Characters with relevant navigation skills handle these challenges more reliably, plotting accurate courses and spotting problems before they become crises.

Random Encounter Table

Roll d12 to determine the random encounter during interplanetary travel:

1. **Gravitational Anomaly**: The spacecraft encounters an unexpected gravitational gradient: a dense cluster, an uncharted mass, or conflicting orbital data. Navigation checks are required to hold course without wasting fuel.
2. **Distress Signal**: The crew picks up a distress signal from another spacecraft or outpost. They can investigate and render assistance or continue on course. The signal may lead to a rescue, a salvage opportunity, or an ambush.

3. **Space Debris**: The spacecraft encounters a cloud of debris: discarded satellites, fragments of destroyed vessels, or industrial waste. Careful maneuvering is required to avoid collisions and hull damage.
4. **Derelict Vessel**: The crew comes across an unresponsive spacecraft broadcasting an old transponder code. Investigation may yield salvageable resources or critical information, or reveal why the ship was abandoned.
5. **Navigation Failure**: A software fault or sensor error causes the spacecraft to veer off the calculated trajectory. The crew must troubleshoot and correct before fuel reserves are committed to the wrong vector.
6. **Asteroid Field**: The spacecraft enters a denser-than-charted region of the Belt or a debris cluster. Navigation skill is needed to thread through without damage.
7. **System Failure**: A cascade fault in a primary ship system (propulsion, life support, or comms) demands immediate attention before it worsens. The crew must diagnose and act fast.
8. **Hostile Intercept**: A vessel orders the ship to heave to. Whether it's a patrol enforcing a blockade, a privateer, or something murkier isn't clear until the crew responds.
9. **Trade Opportunity**: A merchant or supply vessel offers a chance to trade or barter. Valuable resources, spare parts, or information may be available, at a price.
10. **Celestial Event**: A solar flare, meteor shower, or comet flyby creates both a hazard and an observation opportunity. Electronics may be affected; the data collected could be valuable.
11. **Fuel Leak**: A micro-fracture in a fuel line or tank has been bleeding reserves. The crew must assess the loss, repair what they can, and decide whether to push on or divert.
12. **Unexpected Contact**: A vessel's transponder matches one reported lost or destroyed months ago. Investigation reveals a salvage situation,

a crime scene, or an uncomfortable truth someone would rather stay buried.

Resource Management

Resource management is a crucial aspect of interplanetary travel in “Sol: Beyond Earth.” Characters must carefully monitor and manage their resources to ensure their survival and successful exploration of the solar system. This includes maintaining adequate fuel levels, managing life support systems, and efficiently utilizing consumables necessary for their journey.

Fuel Management: Fuel is the lifeblood of interplanetary travel. Characters must keep a close eye on their spacecraft’s fuel levels and plan their routes accordingly. Each type of propulsion system may have different fuel requirements and consumption rates. Characters must make strategic decisions about refueling, considering available fueling stations, mining opportunities, or potential trade routes for obtaining the necessary resources to continue their journey.

Life Support Systems: Life support systems are vital for sustaining the crew’s survival during the long-duration interplanetary travel. These systems regulate air supply, temperature control, waste management, and other essential functions necessary for human life. Characters must ensure that these systems are properly maintained, and consumables such as oxygen, water, and food are rationed effectively to avoid shortages and maintain crew well-being.

Consumables: In addition to life support resources, characters must manage other consumables required for their journey. This may include spare parts, medical supplies, communication equipment, and scientific instruments. Scarcity of these consumables can create tense situations, forcing characters to prioritize their usage, make repairs when necessary, and potentially engage in trade or negotiations to acquire additional supplies.

Scarcity and Decision-Making: Resources are limited in the vastness of space, and characters will often face scarcity and tough decisions.

They must carefully evaluate their priorities, weighing the need for exploration, survival, and potential rewards. Characters may encounter situations where they must choose between refueling at a distant outpost or conserving fuel for a longer journey, or deciding whether to trade valuable resources for critical supplies or hold onto them for future opportunities.

Random Events and Resource Challenges: Random events, encounters, and navigation challenges can present further tests of resource management. Characters may encounter unexpected resource-rich regions, encounters with desperate survivors, or situations that require sacrifices to ensure the survival of the crew. These events provide opportunities for characters to showcase their resourcefulness, adaptability, and problem-solving skills.

Space Travel Primer

Orbital transfers. Spacecraft don't travel in straight lines. The most fuel-efficient path between two planets is a Hohmann transfer: an elliptical arc that coasts from one orbit to the other using only two engine burns. This makes launch windows critical: planets must be in the right positions relative to each other, or the ship must burn far more fuel to compensate. Miss the window, and the next one may be months or years away.

Changing orbits. Any change in a spacecraft's speed or direction requires an engine burn. Burn in the direction of travel and you rise to a higher orbit; burn against it and you descend. Every maneuver costs fuel and must be calculated precisely; errors compound over millions of kilometers.

Gravity assists. A spacecraft can gain or lose velocity by flying close to a massive body. Threading the right trajectory past a planet or moon transfers momentum from that body's orbit to the ship, or bleeds it off, without using engine fuel. Used correctly, a gravity assist can cut months off a transit or redirect a ship onto a trajectory that would otherwise be prohibitively expensive.

Departure Checklist

Onboard computers handle the mathematics. Before committing to a burn, the crew verifies:

1. Destination and mission objectives confirmed
2. Launch window timing within tolerance
3. Planned trajectory reviewed for course corrections
4. Radiation shielding and micrometeoroid protection assessed
5. Propulsion systems confirmed operational
6. Fuel, life support, and consumables validated for full transit
7. Payload, crew safety, and mission duration constraints checked
8. Contingency procedures and backup systems in place
9. Communications confirmed operational
10. Navigation data current and approved by mission authority

Hazards and Threats

Space travelers face a variety of hazards and threats during their journeys. These can include:

1. **Microgravity:** The absence of gravity in space can have various effects on the human body, such as muscle and bone loss, cardiovascular issues, and fluid distribution problems. Countermeasures such as regular exercise and proper nutrition are essential to mitigate these effects.

2. **Radiation:** Space is filled with various forms of radiation, including cosmic rays and solar radiation. Prolonged exposure to high levels of radiation can be harmful to humans and electronic systems. Shielding and monitoring systems are necessary to protect astronauts and sensitive equipment.

3. **Micrometeoroids and Space Debris:** Small particles and debris floating in space can pose a significant risk to spacecraft and spacewalkers. These objects travel at high speeds and can cause damage upon impact. Protective measures such as shielding and monitoring systems are crucial to minimize the risk.

4. **Extreme Temperatures:** Space experiences extreme temperatures, ranging from intense heat when exposed to direct sunlight to extreme cold in shadowed areas. Spacesuits and spacecraft need to be equipped with insulation and temperature regulation systems to protect astronauts and sensitive equipment.
5. **Life Support System Failure:** Malfunctions or failures in life support systems, including oxygen supply, temperature regulation, and waste management, can jeopardize the health and safety of space travelers. Redundant systems and thorough maintenance procedures are essential to ensure the reliability of life support systems.
6. **Psychological and Social Challenges:** Long-duration space travel and isolation from Earth can have significant psychological and social impacts on astronauts. Issues such as confinement, loneliness, and stress need to be addressed through psychological support systems and regular communication with mission control and loved ones.
7. **Communication Delays:** The vast distances in space can result in significant communication delays between spacecraft and mission control or between different spacecraft. This can pose challenges in emergency situations or when real-time decision-making is necessary.
8. **Navigation and Course Correction:** Navigating in space requires precise calculations and adjustments due to orbital dynamics, gravitational influences, and complex flight paths. Errors in navigation or failure to execute proper course corrections can lead to significant deviations from the intended trajectory.
9. **Fuel and Resource Management:** Spacecraft have limited fuel and resources, and careful management is required to ensure the success of missions. Inefficient resource usage or unexpected events can result in shortages and affect the overall mission objectives.
10. **Human Error:** Mistakes or errors in judgment during space missions can have severe consequences. Proper training, clear procedures, and effective teamwork are crucial to minimize the risk of human error.

Starships

Starships have basic versions of these functions; upgrades cost ₵10 each. In an emergency, players pick an action to perform or *help* with.

COMMS: Radio transmitter. Upgrades: Tighter encryption, faster data burst.

ESCAPE CRAFT: Pod per person. Upgrade: Additional shuttles.

HULL: Aeroshell for re-entry. Upgrades: Reinforced armor, solar shielding.

LIFE SUPPORT: Basic rations and breathing air. Upgrades: Extended mission endurance, hydroponics.

NAVIGATION: Onboard computers. Upgrades: Deep space scanners, autopilot.

POWER: Fuel cells and batteries. Upgrades: RTG or fission generator, more redundancy.

PROPULSION: Main engine and RCS thrusters. Upgrades: Faster acceleration, more delta-V.

WEAPONS: None standard. Upgrades: Point-defense lasers, missile launchers, railguns.

Starship Types

- **Shuttle**: Shuttles are small spacecraft designed for short-distance travel between planetary surfaces and orbital stations. They serve as reliable transportation for crew members, cargo, and supplies, with a focus on versatility and ease of use.

- **Orbital Transfer Vehicle**: Orbital Transfer Vehicles (OTVs) are specialized spacecraft used for transferring payloads, including satellites, modules, and supplies, between different orbits within a planetary system. They possess efficient propulsion systems and precise maneuvering capabilities.
- **Ion Rocket**: Ion Rockets utilize ionized particles as propellant to generate thrust. While their acceleration is relatively low, they provide long-duration, low-thrust propulsion, making them suitable for interplanetary travel and missions requiring efficiency and endurance.
- **Nuclear-Thermal Rocket**: Nuclear-Thermal Rockets use the heat generated by a nuclear reactor to heat propellant and produce thrust. They offer higher thrust levels than ion rockets, enabling faster interplanetary travel. These ships require careful handling of the nuclear components.
- **Mass Driver**: Mass Drivers utilize electromagnetic forces to accelerate payloads to high speeds, making them ideal for launching materials, resources, and even spacecraft into space. They serve as essential infrastructure for interplanetary logistics and resource transport.
- **Solar Sail**: Solar Sail spacecraft utilize the pressure of photons emitted by the sun to propel themselves through space. With vast sail-like structures, they harness the power of sunlight to provide continuous acceleration, allowing for long-distance travel while conserving onboard fuel.
- **Nuclear Pulse Rocket**: Nuclear Pulse Rockets use nuclear explosions to propel the spacecraft forward. By detonating small nuclear charges behind the ship and harnessing the resulting shockwaves, these ships can achieve high speeds and efficient interplanetary travel.
- **Ramjet**: Ramjets use the atmosphere of a planet or moon to collect and compress gas, which is then ignited for propulsion. They are capable of sustained atmospheric flight and offer efficient thrust within planetary atmospheres.

Starship Classes

There can be various ship types or classes, each designed for different purposes and possessing unique attributes and capabilities.

1. **Explorer**: Specialized ships designed for long-range exploration missions, equipped with advanced scanning systems, research labs, and extended fuel capacities.
2. **Freighter**: Cargo ships used for transporting goods, resources, and supplies between planets and stations. They have large cargo holds, efficient propulsion systems, and limited defensive capabilities.
3. **Fighter**: Small and agile combat spacecraft designed for dogfighting and engaging enemy ships in space battles. They prioritize speed, maneuverability, and weapon systems over other attributes.
4. **Bomber**: Heavily armed and armored ships capable of delivering devastating blows to enemy targets, specializing in strategic strikes and bombardment missions.
5. **Cruiser**: Versatile medium-sized ships that balance offensive and defensive capabilities. They are capable of both combat engagements and supporting roles, such as escort missions or long-duration patrols.
6. **Capital Ship**: Massive and heavily armed vessels serving as command centers and flagships for major factions. Capital ships have the highest hull integrity and weapon systems but may lack maneuverability.
7. **Science Vessel**: Advanced research ships equipped with state-of-the-art laboratories and scientific instruments, focused on studying celestial phenomena, conducting experiments, and collecting valuable data.
8. **Mining Ship**: Specialized ships optimized for extracting and processing resources from asteroids, moons, and planetary surfaces. They are equipped with mining equipment, storage facilities, and advanced material processing systems.
9. **Colony Ship**: Purpose-built vessels designed to transport large numbers of people, resources, and equipment for establishing new colonies and settlements on distant planets or moons.

10. **Support Ship**: Utility ships providing various support functions, such as repair, refueling, medical facilities, and cargo handling. They play a crucial role in sustaining operations and maintaining the functionality of other ships and stations.

Space Combat

Conflict Resolution

If conflicts arise between factions during interplanetary travel, space combat mechanics can be used. Combat follows a similar structure to regular combat, with rounds and action sequences.

Ship Attributes

Spacecraft have their own functions, such as Hull, Weapons, Armor, and Propulsion. These determine the capabilities and effectiveness of the spacecraft in combat situations. The Referee can create a set of ship attributes or use pre-defined ship templates for ease of play.

Ship Actions

During space combat, characters can take various ship actions, such as attacking with weapons, evasive maneuvers, deploying countermeasures, or targeting specific ship systems. The actions available depend on the ship's capabilities and the character's skills.

Combat Rolls

Combat rolls are used to determine the outcomes of attacks and defensive actions. These rolls take into account the attacker's skill, ship attributes, and potential modifiers based on the situation. Damage inflicted on the target ship is calculated, considering the defender's armor and other defensive systems.

Boarding Actions

In addition to ship-to-ship combat, characters may engage in boarding actions, attempting to breach enemy ships and engage in close-quarters combat. Boarding actions follow similar rules to regular combat, with characters utilizing their skills and weapons to overcome enemy forces.

Consequences and Repairs

Space combat can have long-lasting consequences, damaging ship systems and potentially causing critical failures. Characters must deal with the aftermath of combat, including repairing their ships, tending to injured crew members, and managing limited resources in the hostile environment of space.

Travel Time and Fuel Consumptions

The time required for interplanetary travel varies depending on the distance between planets and the spacecraft's speed capabilities. Characters should consider travel time when planning their missions and account for potential delays due to unforeseen circumstances or navigation difficulties.

Here's a table showcasing the approximate distances and travel times between major settlements in the Solar System, considering various types of propulsion. Please note that the values provided are estimates and can be adjusted according to the specific needs of your game:

Origin	Destination	Distance (AU)	Travel Time (days) - Ion Rocket	Travel Time (days) - Nuclear-Thermal Rocket	Travel Time (days) - Solar Sail	Travel Time (days) - Nuclear Pulse Rocket
Earth	Lagrange Points	0.01 AU	2	1	1	1
Earth	Luna	0.0026 AU	0.5	0.25	0.25	0.25
Earth	Mars	0.5 AU	100	50	40	60
Earth	Venus	0.28 AU	56	28	20	35
Earth	Belt	2-4 AU	400-800	200-400	150-300	250-500

Origin	Destination	Distance (AU)	Travel Time (days) - Ion Rocket	Travel Time (days) - Nuclear-Thermal Rocket	Travel Time (days) - Solar Sail	Travel Time (days) - Nuclear Pulse Rocket
Earth	Outer Planets	4-30 AU	800-6000	400-3000	300-2000	500-4000
Lagrange Points	Luna	0.0025 AU	0.5	0.25	0.25	0.25
Lagrange Points	Mars	0.49 AU	98	49	39	59
Luna	Mars	0.49 AU	98	49	39	59
Luna	Venus	0.27 AU	54	27	19	34
Luna	Belt	2-4 AU	400-800	200-400	150-300	250-500
Luna	Outer Planets	4-30 AU	800-6000	400-3000	300-2000	500-4000
Mars	Venus	0.76 AU	152	76	57	95
Mars	Belt	1.5-3 AU	300-600	150-300	120-240	200-400
Mars	Outer Planets	3-29 AU	600-5800	300-2900	230-2200	380-3600
Venus	Belt	0.9-2 AU	180-400	90-200	70-150	120-250

Origin	Destination	Distance (AU)	Travel Time (days) - Ion Rocket	Travel Time (days) - Nuclear-Thermal Rocket	Travel Time (days) - Solar Sail	Travel Time (days) - Nuclear Pulse Rocket
Venus	Outer Planets	2-28 AU	400-5600	200-2800	150-2100	250-3400
Belt	Outer Planets	1-26 AU	200-5200	100-2600	80-2000	130-3000

Please note that AU stands for Astronomical Unit, which is the average distance from the Earth to the Sun (approximately 93 million miles or 150 million kilometers).

Fuel consumption will vary depending on the specific propulsion system and spacecraft design. Here's a general guideline for fuel consumption during interplanetary travel:

Propulsion System	Fuel Consumption
Ion Rocket	Low
Nuclear-Thermal Rocket	Moderate
Solar Sail	Negligible
Nuclear Pulse Rocket	High

The fuel consumption can be further adjusted or scaled based on the desired level of realism or gameplay balance. Players will need to manage their fuel resources effectively, refuel at appropriate locations, and plan their journeys accordingly to ensure successful interplanetary travel.

Shipyard

Ulysses - Explorer
HULL - Aeroshell upgrade

ESCAPE CRAFT - 2 pods

NAVIGATION - Deep space scanner upgrade

POWER - Fuel cells

PROPULSION - Main engine, 2x delta-V upgrade

COMMS - Encrypted

Phoenix - Freighter
HULL - Reinforced armor upgrade

ESCAPE CRAFT - 4 pods

CARGO - Heavy loader upgrade

POWER - Fuel cells, redundancy upgrade

PROPULSION - Main engine

NAVIGATION - Autopilot upgrade

Valkyrie - Fighter
HULL - Aeroshell

ESCAPE CRAFT - 1 pod

WEAPONS - 2x laser cannon upgrades

PROPULSION - 2x faster acceleration upgrades

NAVIGATION -

Prometheus - Bomber
HULL - Reinforced armor upgrade

ESCAPE CRAFT - 2 pods

WEAPONS - Missile launcher upgrade

PROPULSION - Main engine

NAVIGATION -

Horizon - Cruiser

HULL - Reinforced armor upgrade

ESCAPE CRAFT - 3 pods

WEAPONS - Point defense lasers

PROPULSION - Main engine, faster acceleration upgrade

NAVIGATION -

Aurora - Science Vessel

HULL - Aeroshell

ESCAPE CRAFT - 2 pods

SENSOR - Survey scanner upgrade

LAB - Chem kit, microscope upgrades

PROPULSION - Main engine

NAVIGATION -

Leviathan - Mining Ship

HULL - Reinforced armor upgrade

ESCAPE CRAFT - 4 pods

EQUIPMENT - Mining tools upgrade

POWER - Fuel cells, redundancy upgrade

PROPULSION - Main engine

Sentinel - Capital Ship

HULL - 2x Reinforced armor upgrades

ESCAPE CRAFT - 5 pods

WEAPONS - 2x missile launcher upgrades

PROPULSION - 2x faster acceleration upgrades

NAVIGATION - Autopilot upgrade

Nebula - Support Ship

HULL - Aeroshell

ESCAPE CRAFT - 2 pods

EQUIPMENT - Repair tools upgrade

PROPULSION - Main engine, delta-V upgrade

NAVIGATION -

Pulsar - Colony Ship

HULL - Reinforced armor upgrade

ESCAPE CRAFT - 8 pods

LIFE SUPPORT - Hydroponics upgrade

PROPULSION - Main engine

NAVIGATION - Autopilot upgrade

Ship Name

1	**Alpha**	**6**	**Bravo**	**11**	**Challenger**	**16**	**Columbia**
2	Discovery	7	Endeavour	12	Enterprise	17	Explorer
3	Freedom	8	Independence	13	Intrepid	18	Liberty
4	Odyssey	8	Pioneer	14	Resolute	19	Spirit
5	Valiant	10	Venture	15	Voyager	20	Zephyr

Environmental Hazards

Space is actively hostile to human survival. The hazards below are constant background conditions that shape every mission, not exceptional events.

Radiation Exposure

Space carries radiation from cosmic sources and solar activity, harmful to both crew and electronics over time. Characters moving through high-radiation regions must use shielding, monitor cumulative exposure, and route around the worst zones where possible. The Referee can introduce radiation events or zones that force decisions about routes, shielding resources, or repair missions on compromised shielding.

Microgravity Environments

In microgravity, weightlessness affects movement, task performance, and equipment handling. Characters need training or tools such as magnetic boots or handheld thrusters to stay stable and work effectively. The Referee can call for skill checks when microgravity makes an action meaningfully harder than it would be on the ground.

Extreme Temperatures

Space runs from near absolute zero in shadow to lethal heat in direct sunlight. Suits and vehicles must maintain thermal regulation; failures cascade quickly into equipment damage or life-threatening conditions. The Referee can introduce temperature events (a failed heater, an exposed suit seam, a vehicle left in direct sun) as complications rather than just background.

Scarcity of Resources

Oxygen, water, food, and power are all finite and must be tracked. Characters who ignore resource levels face degrading options: rationing, diverting from the mission to resupply, or trading away something valuable. The Referee should treat resource pressure as a lever, not just an obstacle.

Environmental Suits and Scientific Equipment

Environmental suits provide protection against the harsh space environments, including temperature extremes, radiation, and micrometeoroids. Characters must have access to and maintain environmental suits to venture outside habitats or spacecraft. Additionally, scientific equipment such as spectrometers, telescopes, or microscopes allow characters to gather valuable data, conduct experiments, and make discoveries. These tools are essential for scientific research and can provide insights into the environment, resources, and potential threats.

Vehicles

Vehicles cost 1₵ to rent for a mission; purchase costs 5₵ unless noted. Damage disables a vehicle until repaired at a base.

- **Lunar Lander (3₵ charter):** A small spacecraft for landing and taking off from the lunar surface. Carries crew and cargo between lunar bases and orbiting spacecraft. Requires a mother ship to deploy; cannot operate independently in transit.
- **Lunar Rover (1₵):** An electric surface vehicle for lunar exploration. Traverses rugged terrain, carries 2 crew and equipment, and can operate for extended periods without refueling. Pressurized cabin optional.
- **Titan Methane Sno-Cat (2₵):** A tracked surface vehicle built for Titan's methane-rich, frigid environment. Insulated against extreme cold, equipped for icy terrain and methane shallows. Used for scientific survey and personnel transport on Titan.
- **Mars Trike (1₵):** A compact three-wheeled surface vehicle for Martian terrain. Solar-charged battery drive. Agile enough for rough ground, carries 1–2 crew with equipment. Not pressurized; suits required.
- **Sub-Orbital Flyer (2₵):** A small winged vehicle for rapid transit within a planetary atmosphere: reconnaissance, atmospheric sampling, or fast passage between surface settlements. Cannot reach orbit under its own power.
- **Lunar Sub-Orbital Bus (2₵):** A pressurized passenger-and-cargo craft for surface-to-orbit trips on Luna. Carries significant payload; serves as the main link between lunar surface settlements and orbital habitats.

- **Lunar Tracked Bus (1₵):** A heavy tracked transport for moving personnel and equipment across the lunar surface. Spacious pressurized cabin; slow but reliable over rough terrain. Standard logistics vehicle for construction and resource operations.
- **Lunar Trailer-Truck (2₵):** A heavy-duty surface hauler consisting of a truck cab and one or more trailers. Used for bulk cargo runs between lunar installations. Durable tires rated for regolith conditions.
- **Ramjet (2₵):** An atmospheric vehicle that collects and compresses ambient gas for thrust, capable of high-speed sustained flight within a thick atmosphere. Used primarily on Venus (floating city transit) and Titan. Cannot operate in vacuum.

The Setting

Timeline

- **1957**: The Soviet Union launches Sputnik 1, the first artificial satellite, marking the beginning of the Space Age.
- **1961**: Yuri Gagarin becomes the first human to orbit Earth aboard Vostok 1.
- **1969**: Apollo 11 lands on Luna; Neil Armstrong becomes the first person to walk on its surface.
- **1971**: The Soviet Union's robotic Mars program achieves its first successful Mars landing.
- **1972**: NASA approves the Lunar Industrial Base initiative, shifting focus from exploration to permanent infrastructure.
- **1978**: First extraction of helium-3 from lunar regolith; fusion energy research accelerates.
- **1981**: The Space Shuttle program launches, establishing a reusable orbital transport capability.
- **1985**: The first human mission lands on Mars (Ares-1): six crew, 18-month stay before withdrawal. The mission proves human Mars operations are feasible but exposes the scale of infrastructure required.
- **1990**: Mars program scaled back; priority shifts to lunar industrialization and asteroid survey missions.
- **1995**: Helium-3 extraction becomes economically viable. Fusion reactors achieve net-positive energy output.

- **1998**: Luna-5 habitat reaches 90% closed-loop life support, the first largely self-sustaining space installation.
- **2003**: Luna-Center permanent base reaches 1,000 residents.
- **2005**: First fusion-powered tug completes an Earth–Mars transit in 90 days, cutting previous travel times in half.
- **2012**: First commercial asteroid mining operation begins processing platinum-group metals in the Belt.
- **2024**: First O'Neill cylinder, "New Shanghai," is completed from lunar materials and orbital fabrication: 10,000 residents.
- **2035**: Mars permanent population reaches 500; first Mars-born child recorded.
- **2040s**: Terraforming research begins on Mars. The process is quickly recognized as a centuries-long engineering program, not a near-term solution.
- **2048**: First Venus floating city, Aphrodite City, established at 50 km altitude: 200 residents.
- **2050s**: Resource extraction from the asteroid belt grows rapidly, establishing Belt-1 and a network of small mining outposts.
- **2065**: First dedicated asteroid belt colony (Belt-1) reaches 500 permanent residents.
- **2072**: Mars population reaches 5,000. First Mars university opens.
- **2075**: Venus floating city population reaches 10,000; first Venus-born child.
- **2082**: First Jupiter moon outpost (Europa-1) establishes a research station beneath the ice shell.
- **2085**: First Saturn moon outpost (Titan-1) begins methane extraction operations.

- **2090**: Total human population beyond Earth reaches approximately 200,000. Tensions between Earth-centered institutions and outer settlements over taxation and resource autonomy come to a head. The Federated Autonomy Treaty establishes the Sol Union: a federation of Earth's major power blocs and outer settlement representatives. The game begins here.

Major Powers

- **European Federation:** Formed by the consolidation of European nations, the European Federation is known for its focus on scientific progress, sustainable development, and cultural diversity. They have a strong presence in both lunar and Martian colonization efforts.
- **North American Block:** Comprising the United States, Canada, and Mexico, the North American Block boasts advanced technology and a pioneering spirit. They have a significant influence on space exploration and maintain several major spaceports.
- **African Union:** United under the African Union, countries in Africa collaborate to participate in space exploration and capitalize on the resources available in the solar system. They focus on sustainable development and resource management.
- **Arabian Coalition**: The Arabian Coalition is composed of Middle Eastern nations that have made significant investments in space exploration, particularly in terraforming efforts on Mars and utilizing space-based solar power.
- **Indian Republic:** India has emerged as a major player in space exploration, driven by its scientific expertise and ambitious space program. They have established a strong presence on Luna and are actively involved in resource extraction.
- **Commonwealth of Independent States:** Formerly part of the Soviet Union, the Commonwealth of Independent States (CIS) continues to pursue space exploration. They have a substantial presence in orbital habitats and focus on scientific research and technological advancements.

- **People's Republic of China:** China has become a key player in space exploration, emphasizing both scientific research and commercial interests. They have a significant presence on Luna, Mars, and the Belt, focusing on resource extraction and industrial development.
- **Pacific Alliance:** Comprising countries in the Asia-Pacific region, the Pacific Alliance prioritizes space tourism and commerce. They have invested in advanced spaceports and orbital infrastructure to facilitate trade and travel.

European Federation

- **Overview:** The European Federation is a major power in the solar system, formed through the consolidation of European nations. It is recognized for its emphasis on scientific progress, sustainable development, and cultural diversity. The Federation has a significant presence in lunar and Martian colonization efforts, contributing to the advancement of humanity's presence in space.
- **Background and History:** The European Federation traces its roots back to the early 21st century when European nations sought closer cooperation and integration. Through diplomatic negotiations and agreements, the Federation was formally established, pooling resources and expertise to drive scientific advancements and space exploration.
- **Leadership and Structure:** The European Federation operates under a decentralized governance structure, with representatives from member nations forming a council to make collective decisions. The council elects a President who serves as the head of state, responsible for leading the Federation's initiatives and representing its interests on the global stage.
- **Territory and Influence:** The European Federation holds influence over various regions within the solar system. It maintains colonies and research stations on Luna and Mars, establishing a significant presence in these key locations. The Federation's territory extends beyond physical settlements, with its cultural influence, scientific collaborations, and economic ties reaching across the solar system.

- **Economic Activities:** The European Federation is actively engaged in a wide range of economic activities within the solar system. These include scientific research, resource extraction, manufacturing, and trade. The Federation promotes sustainable development practices, seeking to balance economic growth with environmental stewardship and social responsibility.
- **Military and Security:** The European Federation maintains a capable and technologically advanced military force to safeguard its interests and ensure the security of its colonies. The military primarily focuses on defensive capabilities and cooperative security agreements with other factions, aiming to maintain peace and stability within the solar system.
- **Science and Technology:** The European Federation places great emphasis on scientific progress and technological advancements. It invests heavily in research and development, particularly in fields such as space exploration, sustainable energy, biotechnology, and artificial intelligence. The Federation's scientific endeavors contribute to advancements in space travel, environmental studies, and the understanding of the universe.
- **Major Projects and Initiatives:** The European Federation is involved in major projects and initiatives aimed at advancing human presence and knowledge in space. These include the establishment and expansion of lunar and Martian colonies, the development of advanced propulsion systems, the pursuit of terraforming technologies, and collaborative efforts with other factions for ambitious exploration missions.

- **Relations with Other Organizations:** The European Federation maintains positive relations with other organizations and major powers in the solar system. It emphasizes diplomacy, cooperation, and collaboration in its interactions. The Federation actively seeks partnerships and joint ventures with other nations and organizations to further scientific research, space exploration, and sustainable development. It is known for its open dialogue, willingness to share knowledge and resources, and commitment to fostering global cooperation in the pursuit of space-related endeavors. The European Federation is seen as a trusted ally and reliable partner, often sought after for its expertise, technological advancements, and diplomatic acumen.
- **Key Figures:** Notable figures within the European Federation include the President, who serves as the head of state and represents the Federation's interests, along with influential scientists, political leaders, and military commanders who contribute to the organization's vision and strategic objectives.
- **Public Perception and Reputation:** The European Federation is known for scientific rigor, sustainable practices, and multilateral diplomacy. Other factions view it as a reliable partner and a counterweight to more commercially aggressive blocs. Its ethical commitments occasionally slow its operations but strengthen its credibility in treaty negotiations.
- **Motivations and Agendas:** The European Federation's motivations revolve around advancing scientific knowledge, promoting sustainable development, and fostering international collaboration. Its long-term agenda includes expanding human presence in space, unlocking the potential of other celestial bodies, and ensuring the responsible and equitable utilization of resources.

North American Block

- **Overview:** The North American Block, consisting of the United States, Canada, and Mexico, holds significant influence in space exploration and maintains several major spaceports. It leads in commercial space ventures, advanced propulsion research, and private-sector partnerships.

- **Background and History:** The North American Block emerged from a history of collaboration and cooperation between the United States, Canada, and Mexico. The three nations recognized their shared values, scientific expertise, and technological capabilities, leading to the formation of a unified entity dedicated to pushing the boundaries of space exploration and scientific discovery.
- **Leadership and Structure:** The North American Block operates under a federal system, with each member nation maintaining its own government and leadership structures. A central governing body, consisting of representatives from each nation, coordinates collaborative initiatives, policy decisions, and resource allocation related to space exploration.
- **Territory and Influence:** The North American Block's influence extends across the solar system. It maintains colonies, research facilities, and spaceports on various celestial bodies, including Luna, Mars, and select asteroids. These settlements serve as strategic bases for exploration, resource extraction, and scientific endeavors.
- **Economic Activities:** The North American Block leads in commercial space technology, manufacturing, and resource extraction. It maintains trade partnerships across factions, often using technology access as a negotiating lever.
- **Military and Security:** The North American Block maintains a formidable military presence to ensure the security of its colonies, research facilities, and spaceports. The military focuses on defensive capabilities, cooperation with allied factions, and safeguarding the Block's interests against potential threats or conflicts.
- **Science and Technology:** The North American Block places a strong emphasis on scientific research and technological development. It invests heavily in cutting-edge research across various disciplines, including space exploration, robotics, AI, biotechnology, and sustainable energy. The Block's scientific achievements contribute to advancements in space travel, resource utilization, and the understanding of the universe.

- **Major Projects and Initiatives:** The North American Block undertakes ambitious projects and initiatives to expand human presence and knowledge in space. These include the establishment and expansion of colonies on celestial bodies, the development of advanced propulsion systems, the deployment of cutting-edge scientific instruments, and collaborations for space exploration missions with other factions.
- **Relations with Other Organizations:** The North American Block values self-sufficiency and independence, but it also recognizes the importance of collaboration and cooperation with other organizations. It maintains strong ties with its allies, particularly the European Federation, with whom it often engages in joint space exploration missions and scientific collaborations. The Block also fosters relationships with private space companies, encouraging entrepreneurship and innovation in the space industry. While it prioritizes its own interests, the North American Block is open to strategic partnerships and alliances that align with its objectives of advancing scientific knowledge, technological progress, and economic growth.
- **Key Figures:** Notable figures within the North American Block include political leaders, scientific visionaries, military commanders, and pioneering astronauts who have made significant contributions to space exploration and the Block's achievements in the solar system.
- **Public Perception and Reputation:** The Block is seen as commercially aggressive and technologically capable. Its spaceports are the busiest in the inner system; its private sector is the most active. Other factions admire the output and distrust the motives.
- **Motivations and Agendas:** The North American Block's motivations revolve around advancing scientific knowledge, technological innovation, and pushing the boundaries of human exploration in space. Its long-term agenda includes expanding human presence on celestial bodies, driving scientific discoveries, fostering international collaboration, and leveraging technology.

African Union

- **Overview:** The African Union represents a collective effort of African nations in the pursuit of space exploration and utilization of solar system resources. United under a common vision, the African Union focuses on sustainable development, resource management, and advancing the interests of African nations in the solar system. They are committed to leveraging space technology and knowledge to benefit their countries and contribute to the collective progress of humanity.
- **Background and History:** The African Union traces its roots to the desire for unity and collaboration among African nations in the exploration of space. Recognizing the potential of space resources and the importance of technological advancements, African countries came together to form the Union. They aimed to pool their expertise, resources, and cultural diversity to foster sustainable development and address the unique challenges and opportunities presented by space exploration.
- **Leadership and Structure:** The African Union operates under a cooperative framework that emphasizes equality, inclusivity, and shared decision-making. It comprises representatives from member nations, who contribute to policy-making, resource allocation, and collaborative initiatives. The Union is led by a rotating presidency, with each member nation taking turns in assuming leadership responsibilities.
- **Territory and Influence:** The African Union maintains a presence in various regions of the solar system. African nations have established colonies, research outposts, and resource extraction facilities on celestial bodies such as Luna, asteroids, and Mars. These locations serve as platforms for scientific research, resource exploitation, and knowledge sharing, contributing to the Union's influence and progress in space exploration.

- **Economic Activities:** The African Union engages in a wide range of economic activities within the solar system. It focuses on sustainable resource management, responsible mining practices, and the development of technologies for renewable energy. African nations participate in trade, manufacturing, and research partnerships, leveraging their unique resources and expertise to contribute to the Union's economic growth and collective well-being.
- **Military and Security:** The African Union maintains a defensive military force to protect its interests and ensure the security of its colonies and resource extraction facilities. The military primarily focuses on safeguarding the Union's territories, preventing conflicts, and providing humanitarian assistance when needed. The Union emphasizes peaceful resolutions, international cooperation, and the prevention of weaponization in space.
- **Science and Technology:** The African Union places a strong emphasis on scientific research, knowledge sharing, and technology development. It invests in fields such as remote sensing, agriculture, health, and climate science to address pressing challenges in African countries. The Union's scientific endeavors contribute to advancements in space exploration, sustainable development, and resource management.
- **Major Projects and Initiatives:** The African Union undertakes major projects and initiatives aligned with its vision and objectives. These include sustainable development programs, satellite launches for communications and Earth observation, collaborations in space research and education, and initiatives to bridge the digital divide and enhance technological capacity in African nations.

- **Relations with Other Organizations:** The African Union actively seeks partnerships and collaborations with other organizations and major powers in the solar system. It aims to leverage shared resources, knowledge, and expertise to accelerate Africa's progress in space exploration and development. The Union maintains cooperative relationships with the European Federation, North American Block, and other major players, with a focus on mutual benefit, knowledge exchange, and capacity-building initiatives. The African Union emphasizes inclusivity and equitable partnerships, striving for fair resource allocation, and ensuring that the benefits of space exploration reach all member nations and their populations.
- **Key Figures:** Notable figures within the African Union include political leaders, scientists, engineers, and cultural ambassadors who play pivotal roles in shaping the Union's agenda and contributing to its progress in space exploration and sustainable development.
- **Public Perception and Reputation:** The African Union is recognized for its sustainable resource practices and commitment to equitable benefit-sharing. It is newer to deep-space operations than the major blocs, and works actively to close that gap through partnerships and education programs.
- **Motivations and Agendas:** The African Union's motivations center around sustainable development, resource management, and the equitable utilization of space resources. Its long-term agenda includes promoting scientific excellence, fostering technological capacity, addressing socioeconomic disparities, and leveraging space technology and knowledge for the benefit of African nations.

Arabian Coalition

- **Overview:** The Arabian Coalition is a collective of Middle Eastern nations that have made significant investments in space exploration and development. Focused on ambitious terraforming projects on Mars and the utilization of space-based solar power, the Coalition aims to leverage space technology for sustainable development, economic growth, and regional cooperation.

- **Background and History:** The Arabian Coalition emerged from a shared vision among Middle Eastern nations to harness the potential of space for their socio-economic advancement. Drawing inspiration from their rich history and cultural heritage, these nations united to pool their resources, scientific expertise, and technological capabilities to pioneer ambitious space projects and contribute to the broader exploration of the solar system.
- **Leadership and Structure:** The Arabian Coalition operates under a cooperative framework that promotes unity, cultural diversity, and regional collaboration. The coalition is led by a rotating presidency, with each member nation taking turns in assuming leadership responsibilities. Representatives from member nations contribute to decision-making, policy formulation, and the allocation of resources for space exploration initiatives.
- **Territory and Influence:** The Arabian Coalition's influence extends across the solar system, with a particular focus on Mars. Member nations have established thriving colonies, research stations, and industrial complexes on the red planet, driving terraforming efforts, resource extraction, and space-based solar power projects. These endeavors contribute to the Coalition's regional influence and elevate its standing in space exploration.
- **Economic Activities:** The Arabian Coalition's economic activities encompass a broad spectrum of industries within the solar system. It focuses on utilizing space-based solar power to meet the energy needs of its member nations and contribute to the global clean energy transition. The Coalition also engages in resource extraction, manufacturing, and technology development, leveraging its unique expertise and resources to fuel economic growth and enhance regional self-sufficiency.
- **Military and Security:** The Arabian Coalition maintains a dedicated defense force to protect its colonies, research facilities, and space-based infrastructure. Its military capabilities prioritize defensive capabilities, regional stability, and cooperation with allied factions. The Coalition actively promotes the demilitarization of space and advocates for peaceful resolutions to conflicts.

- **Science and Technology:** The Arabian Coalition places a strong emphasis on scientific research, technological innovation, and knowledge sharing. It invests in cutting-edge research across various disciplines, including terraforming, renewable energy, biotechnology, and space-based agriculture. The Coalition's scientific achievements contribute to advancements in sustainable development, resource utilization, and environmental stewardship in space and on Earth.
- **Major Projects and Initiatives:** The Arabian Coalition spearheads major projects and initiatives aligned with its vision and objectives. These include ambitious terraforming projects on Mars, the development of space-based solar power infrastructure, collaborations in scientific research, and educational initiatives to inspire and empower the next generation of scientists and engineers in the Middle East.
- **Relations with Other Organizations:** The Arabian Coalition engages in strategic partnerships and collaborations with other organizations and major powers, with a particular emphasis on space-based solar power and terraforming efforts. It seeks to work closely with the European Federation, North American Block, and other leading players to exchange expertise, foster innovation, and drive advancements in sustainable energy solutions. The Coalition actively seeks investment opportunities and joint ventures to maximize its potential in space exploration and commercial ventures. While maintaining its own unique identity and priorities, the Arabian Coalition values cooperation and dialogue to address common challenges and explore new frontiers.
- **Key Figures:** Notable figures within the Arabian Coalition include visionary leaders, prominent scientists, engineers, and cultural ambassadors who play instrumental roles in advancing the Coalition's goals and driving its achievements in space exploration and sustainable development.
- **Public Perception and Reputation:** The Arabian Coalition is known for large, long-horizon investments: Mars terraforming infrastructure and space-based solar power are both generational projects. Other factions respect the ambition and watch the finances carefully.

- **Motivations and Agendas:** The Arabian Coalition's motivations revolve around regional development, sustainability, and advancing the technological capabilities of Middle Eastern nations. Its long-term agenda includes expanding human presence on Mars, driving clean and renewable energy solutions, fostering regional cooperation, and promoting cultural exchange and understanding through space exploration.

Indian Republic

- **Overview:** The Indian Republic has become a prominent player in space exploration, driven by its scientific expertise, technological advancements, and ambitious space program. With a strong presence on Luna and active involvement in resource extraction, India aims to leverage space technology for scientific discovery, economic growth, and national development.
- **Background and History:** The Indian Republic's space program began with the establishment of the Indian Space Research Organization (ISRO) and its ambitious space program. Drawing upon its rich scientific heritage, India has made significant strides in space technology, launching satellites, conducting lunar missions, and establishing colonies on Luna. The nation's commitment to scientific research, innovation, and national progress drives its endeavors in the solar system.
- **Leadership and Structure:** The Indian Republic operates under a democratic framework with a strong emphasis on scientific excellence, collaboration, and public participation. The Indian Space Research Organization (ISRO) plays a central role in the nation's space program, working in close coordination with governmental bodies, research institutions, and private enterprises. The Republic's space policy is guided by the Prime Minister's Office and executed by a dedicated team of scientists, engineers, and administrators.

- **Territory and Influence:** The Indian Republic's influence extends across the solar system, with a notable presence on Luna. Indian lunar colonies serve as centers for scientific research, resource extraction, and technological development. These colonies contribute to the Republic's regional influence and create opportunities for international collaborations in space exploration and utilization.
- **Economic Activities:** The Indian Republic engages in a wide range of economic activities within the solar system. It focuses on resource extraction, scientific research, and the development of space-based technologies. India leverages its expertise in satellite manufacturing, telecommunications, and remote sensing to drive economic growth, enhance national security, and contribute to the global space industry.
- **Military and Security:** The Indian Republic maintains a dedicated space defense force to safeguard its colonies, research facilities, and space-based assets. The force focuses on defensive capabilities, situational awareness, and international cooperation. The Republic places a strong emphasis on maintaining a peaceful and secure environment in space.
- **Science and Technology:** The Indian Republic places great importance on scientific research, technological innovation, and knowledge dissemination. It invests in fields such as remote sensing, astrophysics, planetary science, and space-based applications for agriculture and healthcare. India's scientific endeavors contribute to advancements in space exploration, sustainable development, and societal well-being.
- **Major Projects and Initiatives:** The Indian Republic spearheads major projects and initiatives aligned with its space exploration goals. These include lunar resource extraction programs, satellite launches for communication and Earth observation, missions to explore Mars and other celestial bodies, and initiatives to promote STEM education and scientific awareness among its citizens.

- **Relations with Other Organizations:** The Indian Republic maintains cordial relations with other organizations and major powers in the solar system. It actively seeks collaborative opportunities and partnerships to enhance its space program, share scientific knowledge, and engage in joint missions. The Republic has established cooperative ties with the European Federation, North American Block, and other prominent players, with a focus on technology transfer, capacity building, and scientific exchanges. While pursuing its own national interests, the Indian Republic values mutual respect, shared benefits, and the spirit of exploration that transcends geopolitical boundaries.
- **Key Figures:** Notable figures within the Indian Republic include visionary leaders, distinguished scientists, engineers, and cultural ambassadors who play pivotal roles in shaping the nation's space program, scientific achievements, and international collaborations.
- **Public Perception and Reputation:** India's space program is regarded as one of the most cost-efficient in the system, consistently delivering results at lower per-mission costs than larger blocs. Its lunar presence and satellite networks earn it respect across factions as a reliable technical partner.
- **Motivations and Agendas:** The Indian Republic's motivations center around scientific discovery, national development, and technological self-reliance. Its long-term agenda includes expanding human presence on Luna, driving innovation in space technology, fostering international collaborations, and inspiring future generations to pursue careers in science, technology, engineering, and mathematics (STEM).

Commonwealth of Independent States

- **Overview:** The Commonwealth of Independent States (CIS) emerged from the dissolution of the Soviet Union and remains a formidable player in space exploration. With a significant presence in orbital habitats and a strong focus on scientific research and technological advancements, the CIS continues its spacefaring legacy and contributes to the exploration and development of the solar system.

- **Background and History:** The CIS traces its roots back to the former Soviet Union's pioneering space program. Building upon this heritage, the member states of the CIS have continued to prioritize space exploration, scientific research, and technological innovation. The CIS represents a collaborative effort among its member nations to jointly pursue space-related endeavors while fostering regional cooperation and unity.
- **Leadership and Structure:** The CIS operates under a cooperative framework, with member nations sharing decision-making responsibilities and pooling their resources for space exploration. A central governing body coordinates the collective efforts of the CIS, facilitating collaboration, technological advancements, and the allocation of resources. Each member nation actively participates in shaping the CIS's space policies, strategic objectives, and scientific pursuits.
- **Territory and Influence:** The CIS's influence extends throughout the solar system, with a particular focus on orbital habitats. These habitats serve as centers for scientific research, technology development, and space-based experiments. The CIS's presence in orbital habitats contributes to its regional influence and fosters collaboration with other factions and major powers in space exploration.
- **Economic Activities:** The CIS engages in diverse economic activities within the solar system, focusing on scientific research, technological development, and resource utilization. Member nations contribute their expertise in various fields, including astrophysics, aerospace engineering, robotics, and materials science. The CIS leverages these capabilities to drive economic growth, technology exports, and advancements in space-related industries.
- **Military and Security:** The CIS maintains a dedicated space defense force to protect its orbital habitats, scientific installations, and space-based assets. The force emphasizes defensive capabilities, early warning systems, and collaborative security initiatives with allied factions. The CIS promotes peaceful coexistence, international cooperation, and adherence to space-related treaties and agreements.

- **Science and Technology:** The CIS places a strong emphasis on scientific research, technological innovation, and knowledge dissemination. It invests in fields such as astrophysics, planetary science, space medicine, and advanced propulsion systems. The CIS's scientific achievements contribute to advancements in space exploration, resource utilization, and technological breakthroughs that benefit both its member nations and the global scientific community.
- **Major Projects and Initiatives:** The CIS spearheads major projects and initiatives aligned with its space exploration goals. These include long-duration space missions, collaborative space telescopes, interplanetary probes, and cutting-edge research in microgravity environments. The CIS's initiatives aim to expand humanity's understanding of the solar system, advanced space technologies, and foster international collaboration.
- **Relations with Other Organizations:** The Commonwealth of Independent States (CIS) maintains cooperative relationships with other organizations and major powers, often drawing on its space heritage and scientific expertise. It actively engages in joint research projects, space station collaborations, and technological partnerships with the European Federation, North American Block, and other key players. The CIS values information sharing, scientific collaboration, and the exchange of best practices to further its space exploration objectives. It seeks to leverage its expertise in space technology, engineering, and scientific research to foster mutually beneficial relationships and contribute to the overall progress of space exploration.
- **Key Figures:** Notable figures within the CIS include visionary leaders, renowned scientists, engineers, and influential policy-makers who shape the organization's strategic direction and drive its scientific achievements. These individuals play crucial roles in fostering collaboration, technological advancements, and international cooperation.
- **Public Perception and Reputation:** The CIS carries the Soviet space legacy and has maintained it. Its orbital habitat expertise is unmatched, and its long-duration mission record is the envy of newer programs. It is seen as a serious scientific partner but a cautious political one.

- **Motivations and Agendas:** The CIS's motivations revolve around scientific discovery, technological innovation, regional cooperation, and the pursuit of national development. Its long-term agenda includes pushing the boundaries of space exploration, fostering economic growth through space-related industries, promoting scientific collaboration, and inspiring future generations to pursue careers in STEM fields.

People's Republic of China

- **Overview:** The People's Republic of China has emerged as a prominent and influential player in space exploration, driven by its commitment to scientific research, technological advancements, and commercial interests. With a significant presence on Luna, Mars, and the Belt, China focuses on resource extraction, industrial development, and pushing the boundaries of human understanding.
- **Background and History:** China's space program began with the establishment of the China National Space Administration (CNSA) and its ambitious space program. Drawing upon its rich cultural heritage and technological advancements, China has made significant strides in space technology, satellite launches, lunar missions, and Mars exploration. The nation's dedication to scientific research, economic growth, and international collaboration drives its endeavors in the solar system.
- **Leadership and Structure:** The People's Republic of China operates under a centralized governance system with a focus on strategic planning, scientific excellence, and national development. The China National Space Administration (CNSA) leads the nation's space program, working in close collaboration with government agencies, research institutes, and commercial entities. China's space policy is guided by the vision of national leaders and executed by a dedicated team of scientists, engineers, and administrators.

- **Territory and Influence:** The People's Republic of China's influence extends across the solar system, with notable presence on Luna, Mars, and the Belt. Chinese lunar colonies, Martian outposts, and Belt mining operations are centers for resource extraction, industrial development, and scientific research. China's territorial presence contributes to its regional influence and creates opportunities for collaborations and partnerships in space exploration and commerce.
- **Economic Activities:** The People's Republic of China engages in diverse economic activities within the solar system, focusing on resource extraction, industrial development, and space-based commerce. China leverages its expertise in manufacturing, advanced technologies, and space systems to drive economic growth, secure access to critical resources, and expand its influence in the global space industry. Chinese companies play a significant role in commercial space ventures, satellite manufacturing, and space-based applications.
- **Military and Security:** The People's Republic of China maintains a dedicated space defense force to safeguard its space assets, colonies, and industrial facilities. The force focuses on defensive capabilities, space situational awareness, and collaborative security initiatives with allied factions. China promotes peaceful coexistence, international cooperation, and the prevention of space-based conflicts.
- **Science and Technology:** The People's Republic of China places great importance on scientific research, technological innovation, and knowledge dissemination. It invests in various fields such as astrophysics, planetary science, space medicine, and advanced propulsion systems. China's scientific endeavors contribute to advancements in space exploration, resource utilization, and technological breakthroughs that benefit both its own nation and the global scientific community.

- **Major Projects and Initiatives:** The People's Republic of China leads major projects and initiatives aligned with its space exploration goals. These include lunar resource extraction programs, Mars exploration missions, satellite launches for communication and Earth observation, and initiatives to advance space-based technologies and commercial ventures. China's initiatives aim to enhance scientific knowledge, drive economic growth, and inspire future generations to pursue careers in science, technology, engineering, and mathematics (STEM).
- **Relations with Other Organizations:** The People's Republic of China pursues cooperative relationships and strategic partnerships with other organizations and major powers in the solar system. It actively engages in joint missions, scientific collaborations, and technology sharing initiatives with the European Federation, North American Block, and other leading players. The Chinese space program emphasizes mutually beneficial exchanges, technological advancements, and the peaceful exploration of space. While maintaining its own distinct objectives and interests, China values cooperation, diplomacy, and the spirit of shared exploration to advance scientific knowledge and contribute to the collective progress of humanity in the solar system.
- **Key Figures:** Notable figures within the People's Republic of China include visionary leaders, renowned scientists, influential policy-makers, and successful entrepreneurs who contribute to shaping the nation's space program, scientific achievements, and international collaborations. These individuals play critical roles in advancing China's space agenda and driving technological innovation.
- **Public Perception and Reputation:** China's presence on Luna, Mars, and in the Belt makes it one of the most operationally extensive blocs in the system. Other factions respect the scale and speed of its expansion; some are wary of how quickly it moves to secure strategic sites.

- **Motivations and Agendas:** The People's Republic of China's motivations revolve around scientific discovery, economic growth, and national development. Its long-term agenda includes establishing a sustainable presence on Luna, Mars, and the Belt, fostering international collaborations, driving innovation in space technology, and inspiring future generations to pursue careers in STEM fields.

Pacific Alliance

- **Overview:** The Pacific Alliance is a coalition of countries in the Asia-Pacific region that have come together to promote space tourism, commercial ventures, and economic growth. With a focus on developing advanced spaceports and orbital infrastructure, the Pacific Alliance seeks to facilitate trade, travel, and technological advancements within the solar system.
- **Background and History:** The Pacific Alliance emerged from a shared vision among its member countries to harness the potential of space for economic development and international collaboration. Leveraging their technological expertise, economic strength, and cultural diversity, these nations have joined forces to create a unified front in the global space industry. The Pacific Alliance's history is marked by ambitious space tourism initiatives, investment in spaceports, and a commitment to fostering commercial partnerships.
- **Leadership and Structure:** The Pacific Alliance operates through a cooperative framework, with member countries contributing their expertise, resources, and leadership. A governing body composed of representatives from each member nation oversees the alliance's activities and strategic direction. Decision-making is collaborative, with an emphasis on inclusivity, consensus-building, and equitable distribution of benefits among member countries.

- **Territory and Influence:** The Pacific Alliance's influence extends across the Asia-Pacific region and beyond. Member countries have established advanced spaceports, orbital habitats, and commercial space facilities to serve as hubs for space tourism, trade, and research. The alliance's territorial presence facilitates regional cooperation, economic growth, and technological advancements, attracting both public and private entities to participate in the burgeoning space industry.
- **Economic Activities:** The Pacific Alliance focuses on stimulating economic growth through space tourism, commerce, and technological advancements. Member countries invest in the development of spaceports, launch facilities, and space-related industries to attract both domestic and international businesses. The alliance encourages entrepreneurship, innovation, and collaboration in areas such as satellite manufacturing, space-based applications, and space tourism services.
- **Military and Security:** The Pacific Alliance places emphasis on the peaceful and cooperative use of space, prioritizing international security cooperation and the prevention of conflicts in space. While the alliance may maintain a defense capability to safeguard its assets, it primarily focuses on collaborative security initiatives, information sharing, and promoting transparency among member nations.
- **Science and Technology:** The Pacific Alliance recognizes the importance of scientific research and technological advancements in driving space exploration and economic growth. Member countries invest in research institutions, academic collaborations, and technology transfer programs to advance knowledge in areas such as astrophysics, planetary science, and space medicine. The alliance encourages open data sharing, joint research projects, and the development of cutting-edge technologies.

- **Major Projects and Initiatives:** The Pacific Alliance leads major projects and initiatives aimed at promoting space tourism, commercial partnerships, and technological innovation. These include the establishment of spaceports equipped with state-of-the-art facilities, development of space tourism infrastructure, launch services for satellites and payloads, and initiatives to foster international collaboration in space-related industries.
- **Relations with Other Organizations:** The Pacific Alliance seeks to establish strong relationships with other organizations and major powers, with a primary focus on space tourism and commerce. It actively engages in partnerships and collaborations that promote trade, travel, and the exchange of ideas. The Alliance values open markets, free trade, and the development of international regulations to facilitate space tourism and commercial activities. It maintains cooperative ties with other major players in the solar system, particularly in the areas of spaceports, logistics, and orbital infrastructure. The Pacific Alliance is seen as an influential force in shaping the future of space commerce, attracting investors, and fostering international cooperation in the realm of space tourism and industry.
- **Key Figures:** Notable figures within the Pacific Alliance include visionary leaders, influential entrepreneurs, renowned scientists, and policymakers who contribute to shaping the alliance's strategic direction, economic initiatives, and technological advancements. These individuals play critical roles in fostering collaboration, facilitating international partnerships, and driving innovation within the alliance.
- **Public Perception and Reputation:** The Pacific Alliance runs the busiest commercial ports and the most accessible tourist routes in the system. It is seen as open, pragmatic, and primarily interested in revenue. Political entanglements are avoided where possible; commercial ones are pursued aggressively.
- **Motivations and Agendas:** The Pacific Alliance's agenda centers on maintaining its position as the primary commercial hub of the solar system: spaceport traffic, cargo contracts, tourism revenue, and the regulatory frameworks that protect those interests.

Corporations

- **Kyushucorp:** A Japanese corporation specializing in robotics and artificial intelligence, Kyushucorp provides cutting-edge technology for various space missions, including advanced robotic exploration and maintenance systems.
- **Pizarro Incorporated:** A multinational mining corporation, Pizarro Incorporated operates across the solar system, extracting valuable resources from asteroids, moons, and planets. They are known for their efficient mining operations and resource management.
- **Kozosoft:** A software development company, Kozosoft provides advanced AI systems, virtual reality interfaces, and simulation software for space missions, covering crew training, mission planning, and spacecraft operations.
- **AVQ:** An aerospace vehicle manufacturer, AVQ produces a wide range of spacecraft, including interplanetary vessels, lunar shuttles, and orbital habitats. They are known for their reliable and retrofuturistic designs.
- **Earth Global:** An environmental organization, Earth Global is dedicated to preserving and restoring Earth's ecosystems. They play a significant role in sustainable resource management, advocating for responsible practices in space colonization.

Kyushucorp

Overview: Kyushucorp is a Japanese corporation at the forefront of robotics and artificial intelligence technologies. With a strong presence in the space industry, Kyushucorp plays a crucial role in providing cutting-edge solutions for various space missions. They specialize in the development of advanced robotic systems for exploration, maintenance, and research purposes. Their expertise in robotics and AI has positioned them as a key player in the advancement of space technology.

Public Perception and Reputation: Kyushucorp is highly regarded for its technological prowess and innovative solutions in the field of robotics. They have earned a reputation for reliability, efficiency, and precision in their robotic systems, making them a trusted partner in space exploration.

Their commitment to pushing the boundaries of robotics and artificial intelligence has garnered respect and admiration from both the scientific community and the general public.

Areas of Expertise: Kyushucorp's primary areas of expertise lie in robotics and artificial intelligence. They excel in developing advanced robotic systems specifically designed for space missions. Their technologies enable autonomous exploration, remote operations, and intricate maintenance tasks in challenging space environments.

Notable Products or Services: Kyushucorp offers a range of cutting-edge robotic systems tailored for space exploration. This includes robotic rovers for planetary exploration, maintenance robots for space station upkeep, and autonomous drones for surveying and mapping. They also provide sophisticated AI systems that facilitate data analysis, decision-making, and adaptive learning for space missions.

Relations with Other Organizations: Kyushucorp maintains collaborative partnerships with space agencies, research institutions, and other corporations involved in space exploration. They actively collaborate on joint projects, sharing their expertise in robotics and AI to enhance mission capabilities. Kyushucorp's dedication to collaboration and knowledge exchange has fostered positive relationships within the space community.

Corporate Culture: Kyushucorp values innovation, efficiency, and a strong focus on technological advancements. They foster a culture of collaboration and continuous improvement, encouraging employees to push the boundaries of robotics and AI. The corporation promotes diversity, inclusivity, and the nurturing of talent, attracting top professionals from various fields to contribute to their cutting-edge projects.

Future Goals and Projects: Kyushucorp aims to further advance the field of robotics and AI in space exploration. They are actively involved in the development of next-generation robotic systems capable of autonomous decision-making, adaptive learning, and enhanced mobility. Kyushucorp envisions a future where their technologies revolutionize space missions, enabling humans to explore and understand the cosmos more effectively.

Pizarro Incorporated

Overview: Pizarro Incorporated is a multinational mining corporation operating across the solar system. With a strong focus on resource extraction, Pizarro Incorporated plays a vital role in sourcing and supplying valuable resources from asteroids, moons, and planets. Their expertise in efficient mining operations and resource management has established them as a leading player in the industry.

Public Perception and Reputation: Pizarro Incorporated is widely recognized for its proficiency in resource extraction and its ability to maximize yields from mining operations. They have earned a reputation for their technological advancements in mining techniques and their commitment to responsible resource management. However, their aggressive expansion and resource-centric approach have also attracted criticism from environmental groups concerned about the long-term impact of their operations.

Areas of Expertise: Pizarro Incorporated specializes in mining operations and resource management. They possess advanced technologies and methods for extracting and processing valuable minerals and resources from celestial bodies. Their expertise includes asteroid mining, lunar and planetary resource extraction, and logistical operations associated with the transport and distribution of resources.

Notable Products or Services: Pizarro Incorporated provides a wide range of mining services, including prospecting, drilling, excavation, and mineral processing. They offer comprehensive resource management solutions, including analysis, assessment, and optimization of mining operations. Pizarro Incorporated is known for its efficient extraction techniques, advanced machinery, and sustainable mining practices.

Relations with Other Organizations: Pizarro Incorporated maintains both collaborative and competitive relationships with various organizations. They often partner with space agencies and research institutions to leverage scientific knowledge and exploration data. However, their aggressive expansion and resource-focused approach sometimes lead to conflicts of interest with environmental organizations and indigenous communities concerned about the ecological impact of mining activities.

Corporate Culture: Pizarro Incorporated has a profit-driven corporate culture, prioritizing efficiency, innovation, and financial success. They are committed to technological advancements in mining operations and constantly strive to optimize resource extraction processes. While they prioritize profitability, they also maintain a focus on responsible resource management, implementing sustainable practices wherever possible.

Future Goals and Projects: Pizarro Incorporated aims to expand its mining operations across the solar system, identifying new resource-rich locations and implementing state-of-the-art mining technologies. They seek to develop sustainable mining practices and establish long-term partnerships with organizations and communities to ensure the responsible and ethical extraction of resources. Pizarro Incorporated is also actively involved in research and development efforts to improve resource processing techniques and explore innovative ways to utilize resources in space exploration and colonization endeavors.

Kozosoft

Overview: Kozosoft is a leading software development company specializing in advanced AI systems, virtual reality interfaces, and simulation software for space missions. They provide cutting-edge technology solutions that enhance astronaut training, facilitate mission planning, and optimize spacecraft operations. Kozosoft's expertise in software development has positioned them as a key player in the space industry.

Public Perception and Reputation: Kozosoft's simulation and training platforms are used by most major space agencies and several independent operators. Its software sets the de facto standard for crew certification; anything else requires explicit justification to regulators.

Areas of Expertise: Kozosoft specializes in software development, with a focus on AI systems, virtual reality interfaces, and simulation software for space missions. They excel in creating realistic training environments, advanced data analysis tools, and sophisticated user interfaces that optimize spacecraft operations and enhance decision-making capabilities.

Notable Products or Services: Kozosoft offers a range of advanced software products and services. These include AI systems that support autonomous spacecraft operations, virtual reality interfaces for realistic training scenarios, and simulation software for mission planning and optimization. They also provide data analysis tools that enable efficient processing and interpretation of mission data.

Relations with Other Organizations: Kozosoft collaborates closely with space agencies, research institutions, and spacecraft manufacturers to integrate their software solutions into various space missions. They work hand-in-hand with other organizations to ensure seamless compatibility and the efficient use of their software in different spacecraft systems. Kozosoft's partnerships and collaborations contribute to the overall success and safety of space missions.

Corporate Culture: Kozosoft fosters a culture of innovation, collaboration, and excellence in software development. They encourage creativity and continuous improvement, striving to create cutting-edge solutions that meet the unique needs of space exploration. The company promotes an inclusive and supportive work environment, attracting top talent in software development and cultivating a passion for pushing the boundaries of technology.

Future Goals and Projects: Kozosoft is developing next-generation decision-support AI for real-time mission operations, aiming to reduce crew cognitive load during high-stakes maneuvers. It is also expanding its simulation platform to cover Belt and outer-planet environments, where its current training scenarios are thin.

AVQ

Overview: AVQ is an aerospace vehicle manufacturer known for interplanetary vessels, lunar shuttles, and orbital habitats. Its designs prioritize reliability and maintainability in the field, with a retrofuturistic aesthetic that has become something of a brand identity.

Public Perception and Reputation: AVQ is highly regarded for its commitment to quality and reliability in spacecraft manufacturing. They have earned a reputation for their retrofuturistic designs that blend advanced technology with a touch of nostalgia. AVQ's spacecraft are trusted for their durability, efficiency, and innovative features. Their emphasis on safety and performance has solidified their position as a trusted name in the aerospace industry.

Areas of Expertise: AVQ specializes in the design, manufacturing, and assembly of aerospace vehicles. They have extensive expertise in building interplanetary vessels capable of long-distance travel, lunar shuttles for transportation between Luna and Earth, and orbital habitats to support human habitation in space. AVQ's focus on advanced propulsion systems, life support technologies, and ergonomic spacecraft interiors sets them apart in the industry.

Notable Products or Services: AVQ offers a range of spacecraft tailored to different purposes. Their interplanetary vessels are designed to withstand the harsh conditions of deep space travel, equipped with advanced propulsion systems and state-of-the-art navigation capabilities. Lunar shuttles provide reliable and efficient transportation to and from Luna, enabling lunar exploration and resource extraction. AVQ's orbital habitats offer comfortable living environments with sustainable life support systems for extended stays in space.

Relations with Other Organizations: AVQ collaborates closely with space agencies, research institutions, and other corporations in the aerospace industry. They often partner with scientific missions, providing spacecraft and support systems. AVQ's dedication to reliable and high-quality spacecraft has established strong working relationships with various organizations involved in space exploration and colonization.

Corporate Culture: AVQ fosters a culture of innovation, precision engineering, and attention to detail. They value the expertise and creativity of their engineers and designers, encouraging them to push the boundaries of spacecraft design and functionality. AVQ prioritizes safety, performance, and customer satisfaction, ensuring that each spacecraft they produce meets the highest standards of quality.

Future Goals and Projects: AVQ is dedicated to advancing space exploration and colonization efforts. They continuously seek to refine their spacecraft designs, incorporating new technologies and materials to enhance performance and sustainability. AVQ is actively involved in research and development projects focused on improving propulsion systems, life support technologies, and spacecraft ergonomics. They aim to play a key role in enabling humanity's expansion into space and the successful establishment of sustainable off-world colonies.

Earth Global

Overview: Earth Global is an environmental organization dedicated to preserving and restoring Earth's ecosystems. While their primary focus is on Earth, they also play a significant role in sustainable resource management and advocating for responsible practices in space colonization. Earth Global's mission is to ensure the long-term viability and health of the planet and the solar system as a whole.

Public Perception and Reputation: Earth Global is highly respected for its commitment to environmental preservation and sustainability. They are seen as a champion for Earth's ecosystems and a leading voice in advocating for responsible practices in space colonization. Their efforts to raise awareness about the importance of environmental protection have earned them a positive reputation. Earth Global is known for their scientific expertise, integrity, and dedication to creating a sustainable future for all.

Areas of Expertise: Earth Global specializes in environmental research, resource management, and sustainable development. They have extensive knowledge in assessing the impact of human activities on ecosystems and designing strategies to mitigate environmental harm. Earth Global's expertise extends to space colonization, where they provide guidance on minimizing the ecological footprint of human presence and ensuring responsible resource utilization.

Notable Projects and Initiatives: Earth Global has undertaken various projects and initiatives to protect Earth's ecosystems and promote sustainable practices in space colonization. They conduct research on environmental impacts, develop guidelines for responsible resource extraction in space, and collaborate with space agencies and corporations to integrate sustainability principles into their operations. Earth Global also engages in public outreach and education programs to raise awareness about the importance of environmental stewardship.

Relations with Other Organizations: Earth Global collaborates with space agencies, corporations, and other environmental organizations to promote sustainable practices in space colonization. They work closely with space exploration companies to develop environmentally friendly technologies and protocols. Earth Global also engages in partnerships with other organizations to conduct joint research projects and share best practices in environmental preservation.

Corporate Culture: Earth Global nurtures a culture of scientific rigor, environmental consciousness, and collaboration. They value interdisciplinary approaches to problem-solving and encourage the exchange of ideas and knowledge among their team members. Earth Global fosters a sense of responsibility and urgency in addressing environmental challenges, driven by a shared commitment to safeguarding the planet's ecosystems and promoting sustainable practices.

Future Goals and Projects: Earth Global's future goals revolve around expanding their influence in space colonization and ensuring the integration of sustainable practices in all aspects of off-world endeavors. They aim to develop comprehensive frameworks for ecological impact assessment and resource management in space. Earth Global envisions a future where human activities beyond Earth's boundaries are guided by the principles of environmental sustainability, safeguarding both our home planet and the celestial bodies we explore.

Planets and Other Celestial Bodies

- **Earth:** The administrative hub of the Sol Union, Earth remains the center of political and economic power. It houses numerous orbital stations that serve as transport hubs, communication centers, and research platforms.
- **Luna:** Luna has evolved into a major hub for space activities. It hosts multiple lunar bases and serves as a gateway to the rest of the solar system. Lunar colonies focus on scientific research, resource extraction, and space tourism.
- **Mars:** Terraforming operations are underway but will take centuries to meaningfully alter the atmosphere. Mars hosts thousands of settlers in pressurized habitats, research facilities, and industrial operations, sustained by hydroponics and regular supply runs from Earth.
- **Venus:** Although the surface of Venus remains inhospitable, floating cities in the planet's upper atmosphere have been established. These cities utilize advanced technology to withstand extreme conditions and conduct research on the planet's atmosphere.
- **Belt:** The asteroid belt between Mars and Jupiter is a rich source of valuable minerals and resources. Mining colonies and outposts are scattered throughout this region, extracting resources to support the solar system's growing population.
- **Outer Planets:** The moons of Jupiter and Saturn host research stations, mining colonies, and outposts. These locations focus on scientific exploration, resource extraction, and studying the potential for habitable environments on distant celestial bodies.

Earth

Physical Characteristics:

- **Size and Composition:** Earth is a rocky planet with a diameter of approximately 12,742 kilometers.
- **Atmosphere:** Earth's atmosphere is predominantly composed of nitrogen (78%) and oxygen (21%), with trace amounts of other gases. The atmospheric pressure at sea level is about 101.3 kilopascals.

- **Surface Features:** Earth boasts diverse surface features, including vast oceans, towering mountain ranges, expansive plains, and deep valleys. It also has polar ice caps, deserts, and lush forests.
- **Climate:** Earth experiences a range of climates due to its diverse geography. It has tropical, temperate, and polar regions, with variations in temperature, precipitation, and seasons.
- **Natural Resources:** Earth is rich in various natural resources, including minerals (such as iron, copper, and gold), fossil fuels (such as oil, coal, and natural gas), freshwater sources, and fertile land for agriculture.

Colonization and Settlements:

- **Major Settlements:** Earth is home to the administrative center of the Sol Union, where the newly established federated government is based. It houses various administrative centers, cultural hubs, and centers of innovation. Major settlements also include sprawling megacities, research complexes, and industrial centers.
- **Key Industries:** Earth's key industries revolve around governance, finance, technology development, manufacturing, entertainment, and scientific research. It serves as the central hub for interplanetary trade and resource distribution.
- **Faction Influence:** The major power blocs (the European Federation, North American Block, African Union, Arabian Coalition, Indian Republic, Commonwealth of Independent States, People's Republic of China, and Pacific Alliance) are the founding members of the Sol Union, and they exert their influence and compete for resources and political power on Earth and beyond.

Hazards and Challenges:

- **Environmental Hazards:** While Earth is the most habitable planet in the solar system, it still poses hazards such as natural disasters (hurricanes, earthquakes, etc.), pollution, and climate change due to human activities.

- **Navigation Challenges:** Earth's complex orbital dynamics require precise navigation calculations for spacecraft traveling to and from the planet. This includes coordinating with orbital stations, avoiding space debris, and managing re-entry into the atmosphere.

Points of Interest:

- **Notable Landmarks:** Earth boasts iconic landmarks, including natural wonders like the Grand Canyon, the Great Barrier Reef, and Mount Everest, as well as human-made structures like the Eiffel Tower, the Great Wall of China, and the Pyramids of Giza.
- **Research Opportunities:** Earth presents countless research opportunities in various fields, such as climate studies, biodiversity conservation, urban planning, sustainable energy solutions, and social sciences.
- **Hidden Secrets:** Beneath Earth's vast oceans and unexplored territories lie hidden secrets, such as undiscovered archaeological sites, hidden ecosystems, and geological phenomena yet to be fully understood.

Travel and Logistics:

- **Interplanetary Travel:** Traveling from Earth to other major settlements within the solar system requires advanced interplanetary spacecraft. The travel time and distance vary depending on the destination and the type of propulsion used.
- **Supply and Trade:** Earth serves as a central hub for supply distribution, trade networks, and economic activities. It receives resources from other settlements and distributes them among various factions, ensuring the survival and prosperity of the solar system's human population.
- **Infrastructure and Facilities:** Earth's infrastructure includes orbital stations, spaceports, research institutions, transportation networks, communication centers, and advanced manufacturing facilities. These facilities support interplanetary travel, scientific endeavors, and the smooth functioning of the global government.

As the administrative hub of the Sol Union, Earth is where political authority concentrates, trade routes converge, and the tensions between Earth-based institutions and outer settlements play out most visibly.

Luna

Physical Characteristics:

- **Size and Composition:** Luna is Earth's only natural satellite, with a diameter of approximately 3,474 kilometers. It is primarily composed of rocky terrain, covered with regolith (a layer of loose soil and rocky fragments).
- **Atmosphere:** Luna has an extremely thin and almost non-existent atmosphere, known as an exosphere. It consists of trace amounts of various gasses, including helium, neon, and hydrogen, but it is not capable of supporting human life without artificial life support systems.
- **Surface Features:** Luna's surface is marked by vast plains known as maria, formed by ancient volcanic activity. It also features numerous craters of different sizes, mountain ranges, and deep valleys called rilles.
- **Climate:** Luna has an extreme climate, with significant temperature variations. During the lunar day (approximately 14 Earth days), surface temperatures can reach up to 127 degrees Celsius (261 degrees Fahrenheit), while during the lunar night, temperatures can drop to around −173 degrees Celsius (-279 degrees Fahrenheit).
- **Natural Resources:** Luna possesses valuable resources such as Helium-3, a potential fuel source for fusion reactors. It also contains other minerals, metals, and rare earth elements that can be extracted and utilized for various purposes.

Colonization and Settlements:

- **Major Settlements:** Luna is home to multiple lunar bases and settlements established by various nations and private corporations. These settlements serve as research outposts, resource extraction centers, and launch facilities for interplanetary missions.
- **Key Industries:** Luna's key industries revolve around scientific research, resource extraction, and space tourism. Scientists conduct studies on lunar geology, astrophysics, and biology. Resource extraction operations focus on mining Helium-3 and other valuable materials. Space tourism companies offer visitors the opportunity to experience lunar landscapes and engage in lunar activities.

- **Faction Influence:** Major Sol Union power blocs and corporations have established their presence on Luna, with each vying for control over resource-rich areas and strategic locations. The European Federation, North American Block, People's Republic of China, and various private enterprises all maintain installations and competing interests in lunar settlements.

Hazards and Challenges:

- **Environmental Hazards:** Luna's environment is microgravity, extreme temperature swings, and no atmosphere. Characters operating outside must wear environmental suits with functional life support at all times.
- **Navigation Challenges:** Precise navigation is crucial for lunar travel to avoid hazardous areas, locate resource-rich regions, and safely land or take off from the moon's surface. Characters may encounter unexpected lunar surface conditions, such as unstable terrain or hazardous regolith pockets.

Points of Interest:

- **Lunar Bases:** Highlight major lunar bases where characters can interact with scientists, engineers, and personnel involved in research and resource extraction. These bases may vary in size and purpose, ranging from smaller research outposts to larger settlements with advanced infrastructure.
- **Resource Extraction Sites:** Identify areas on Luna where resource extraction operations are concentrated. These sites may feature mining facilities, processing plants, and storage facilities for the extracted resources.
- **Tourist Attractions:** Lunar settlements offer unique experiences for tourists, including guided excursions to iconic lunar landmarks, low-gravity sports activities, and breathtaking views of Earth from the moon's surface.
- **Scientific Research Stations:** Mention scientific research stations where characters can engage in cutting-edge research across various disciplines, collaborate with scientists, or undertake missions related to lunar exploration and understanding.

Travel and Logistics:

- **Interplanetary Travel:** Luna serves as a gateway to the rest of the solar system, with regular transport options available for interplanetary travel. Travel time from Earth to Luna can vary depending on the propulsion systems used, but it typically takes a few days to reach the moon.
- **Supply and Trade:** Luna relies on regular shipments of supplies and equipment from Earth and other settlements within the solar system. Trade networks are established for resource exchange, scientific collaborations, and tourism-related services.
- **Infrastructure and Facilities:** Lunar settlements have developed advanced infrastructure, including habitats, research laboratories, communication networks, launch facilities, and transportation systems (such as lunar rovers or transport shuttles) for efficient movement across the moon's surface.

Luna is the system's most accessible off-world destination and its most developed: research stations, mining operations, and tourist facilities within a few days' transit from Earth.

Mars

Physical Characteristics:

- **Size and Composition:** Mars is a terrestrial planet with a diameter of approximately 6,779 kilometers, about half the size of Earth. It has a rocky composition, including iron-rich soil and surface features.
- **Atmosphere:** Mars has a thin atmosphere composed primarily of carbon dioxide (95%) with traces of nitrogen and argon. The atmospheric pressure on Mars is about 0.6% of Earth's atmospheric pressure.
- **Surface Features:** Mars is known for its distinct surface features, including expansive deserts, vast valleys, towering volcanoes, and a polar ice cap made up of frozen water and carbon dioxide.
- **Climate:** Mars has a cold and dry climate, with average surface temperatures ranging from −87 degrees Celsius (-125 degrees Fahrenheit) in the winter to −5 degrees Celsius (23 degrees Fahrenheit) in the summer. It experiences seasonal variations and occasional dust storms.

- **Natural Resources:** Mars holds various natural resources, including water ice in its polar regions, underground water reservoirs, minerals such as iron, aluminum, and silicon, as well as potential underground geothermal energy sources.

Colonization and Settlements:

- **Major Settlements:** Mars hosts a network of pressurized habitat clusters, research stations, and early industrial facilities established by different factions and corporations. The largest settlements number in the hundreds to low thousands of residents. Infrastructure is growing fast but remains sparse by Earth standards.
- **Key Industries:** Martian operations focus on resource extraction (mining minerals and accessing underground water ice) and the scientific work required to advance terraforming. Hydroponics and controlled environment agriculture support local food production. Everything else still arrives by supply run.
- **Faction Influence:** The European Federation, North American Block, People's Republic of China, and Arabian Coalition all maintain facilities on Mars, competing for access to the best terraforming sites, mineral deposits, and strategic high ground. A small but growing population of Mars-born residents is beginning to assert its own political identity.

Hazards and Challenges:

- **Environmental Hazards:** Mars poses challenges such as low gravity, extreme temperatures, thin atmosphere, and the potential for dust storms. Characters must adapt to these conditions by wearing specialized suits, managing life support systems, and protecting against radiation exposure.
- **Navigation Challenges:** Navigating the Martian surface requires careful consideration of topography, potential hazards (such as steep cliffs or sand dunes), and the efficient use of rovers or other transportation methods.

Points of Interest:

- **Tharsis Region:** Highlight the Tharsis region, home to the tallest volcano in the solar system, Olympus Mons, and other notable volcanic formations. This region offers geological and scientific research opportunities.
- **Valles Marineris:** Mention Valles Marineris, a vast canyon system on Mars, which presents exploration and scientific research prospects due to its unique geological formations and potential for understanding Martian history.
- **Polar Ice Caps:** Highlight the polar ice caps of Mars, where frozen water and carbon dioxide exist. These regions offer potential water extraction sites and valuable resources for supporting human settlements.
- **Research Facilities:** Identify research facilities and laboratories where characters can engage in cutting-edge scientific studies, explore Martian geology, study astrobiology, or conduct experiments in controlled environments.

Travel and Logistics:

- **Interplanetary Travel:** Traveling from Earth to Mars requires advanced interplanetary spacecraft and can take several months, depending on the alignment of the planets and the chosen propulsion system.
- **Supply and Trade:** Mars relies on regular shipments of supplies from Earth and other settlements for sustenance, equipment, and technological advancements. Trade networks are established for resource exchange, scientific collaborations, and the importation of specialized goods.
- **Infrastructure and Facilities:** Martian settlements have developed extensive infrastructure, including pressurized habitats, transportation networks, resource extraction facilities, research laboratories, and energy generation systems (such as solar panels or potential geothermal power).

In *Sol: Beyond Earth*, Mars is a world in active transformation: pressurized habitats, mining operations, and early terraforming infrastructure create a frontier environment where every resource matters and the next supply ship is never far from anyone's mind.

Venus

Physical Characteristics:

- **Size and Composition:** Venus is similar in size to Earth, with a diameter of approximately 12,104 kilometers. It is a rocky planet with a dense atmosphere primarily composed of carbon dioxide.
- **Atmosphere:** Venus has a thick atmosphere that exerts a surface pressure about 92 times greater than Earth's. It consists mainly of carbon dioxide, with traces of nitrogen and other gasses. The atmosphere also contains clouds of sulfuric acid.
- **Surface Features:** The surface of Venus is covered in vast plains, highland regions, and numerous impact craters. It has volcanic features, including mountains, lava channels, and shield volcanoes.
- **Climate:** Venus has a hostile climate with extreme temperatures and a runaway greenhouse effect. Surface temperatures average around 462 degrees Celsius (864 degrees Fahrenheit), making it the hottest planet in the solar system.
- **Natural Resources:** Venus lacks easily accessible natural resources on its surface due to the harsh conditions. However, its atmosphere contains significant amounts of carbon dioxide, which could potentially be used for fuel or terraforming purposes.

Colonization and Settlements:

- **Floating Cities:** To overcome the inhospitable surface conditions, humanity has established floating cities in Venus's upper atmosphere. These cities rely on advanced technology and engineering to create habitable environments, shielding inhabitants from the extreme temperatures, pressure, and corrosive atmosphere.
- **Research and Exploration:** The floating cities serve as research outposts, studying Venus's atmosphere, weather patterns, and unique geological features. They also support missions to explore nearby cloud layers and conduct experiments to unlock the planet's mysteries.

- **Key Industries:** Industries on Venus focus on atmospheric research, gas extraction, and the development of technologies suited to extreme conditions. Companies and organizations involved in resource extraction and scientific research have a significant presence in the floating cities.

Hazards and Challenges:

- **Extreme Atmospheric Conditions:** The thick atmosphere and intense heat pose challenges to human habitation and operations. Characters must rely on advanced environmental suits and protective equipment to survive and work in the floating cities.
- **Pressure and Gravity:** Venus has a surface gravity about 90% that of Earth's. Adjusting to the different gravitational conditions and managing physical tasks accordingly is essential for characters.

Points of Interest:

- **Floating Cities:** Describe the diverse floating cities in Venus's upper atmosphere, each with its own unique design, architecture, and research focus. These cities house research laboratories, living quarters, advanced life support systems, and atmospheric monitoring stations.
- **Cloud Layers:** Highlight the intriguing cloud layers of Venus, where atmospheric research missions and exploration take place. These layers offer opportunities to study the planet's weather patterns, atmospheric chemistry, and potential signs of microbial life.
- **Aerial Vehicles:** Mention the use of specialized aerial vehicles, such as airships or drones, for transportation, atmospheric sampling, and surveying the planet's features. These vehicles are vital for maneuvering within the dense atmosphere and accessing different regions.
- **Climate-Controlled Facilities:** Identify specialized facilities within the floating cities, such as climate-controlled labs, observation decks with panoramic views of Venus, and research centers dedicated to studying Venus's atmospheric composition and dynamics.

Travel and Logistics:

- **Interplanetary Travel:** Traveling from Earth to Venus requires advanced interplanetary spacecraft and can take several months, depending on the alignment of the planets and the propulsion system used.

- **Supply and Trade:** The floating cities of Venus rely on regular shipments of supplies from Earth and other settlements within the solar system to sustain their inhabitants. Trade networks facilitate the exchange of resources, scientific data, and advanced technologies.
- **Infrastructure and Facilities:** Floating cities require sophisticated infrastructure to provide habitable environments, sustenance, energy generation, and research facilities. These include atmospheric processing systems, advanced life support systems, communication networks, and docking ports for spacecraft.

Venus is one of the system's stranger postings: physically close to Earth but operationally isolated, with no surface access and atmospheric conditions that destroy conventional equipment in hours.

Belt

Overview: The asteroid belt is a vast region located between the orbits of Mars and Jupiter. It consists of numerous small celestial bodies, primarily asteroids, ranging in size from a few meters to hundreds of kilometers in diameter. In "Sol: Beyond Earth," the asteroid belt is a crucial resource hub, with mining colonies and outposts established to extract valuable minerals and resources.

Mining Colonies and Outposts: Mining colonies and outposts are scattered throughout the asteroid belt, each with its own specialization and resource extraction operations. These settlements are often situated on or within larger asteroids, providing a base for mining activities, research, and living quarters for the inhabitants. Some colonies are more developed and self-sustaining, while others rely on regular shipments from other settlements or Earth.

Resource Extraction: The primary focus of the Belt settlements is resource extraction. Asteroids in the belt contain a wide variety of valuable minerals and resources, including metals such as iron, nickel, cobalt, and platinum group elements. Additionally, water ice and organic compounds can be found in some asteroids, providing crucial resources for life support, fuel production, and terraforming efforts.

Mining Operations: Mining operations in the Belt involve various techniques, including drilling, blasting, and excavating asteroids to extract the desired resources. Specialized mining spacecraft, robotic drones, and automated systems are used to perform these tasks. Resource extraction is a challenging endeavor, requiring skilled personnel, advanced technology, and efficient resource management to maximize yields.

Challenges and Hazards: Operating in the asteroid belt presents several practical problems:

- **Navigational Hazards:** Navigating through the asteroid belt requires precise course plotting, avoiding collisions with other asteroids, and mitigating the risk of micro-meteoroid impacts.
- **Limited Resources:** Although the asteroid belt is rich in resources, the logistics of mining and transportation can be complex. Characters must manage resources efficiently, including fuel, water, and life support provisions.
- **Gravity Variations:** The gravitational forces exerted by multiple asteroids in close proximity can cause gravitational anomalies, affecting spacecraft trajectories and requiring careful maneuvering.

Points of Interest:

- **Major Mining Colonies:** Highlight some major mining colonies or outposts in the asteroid belt, each with its unique characteristics, resource specialization, and political affiliations. These settlements can serve as hubs for trade, research, and interactions with other factions in the solar system.
- **Rare Mineral Deposits:** Identify specific asteroids or regions within the belt that harbor rare or highly valuable mineral deposits. These areas might attract intense competition among mining operations and spark conflicts over resource ownership.
- **Research Facilities:** Mention research facilities located within the asteroid belt, where scientists study asteroid composition, conduct experiments in microgravity environments, and develop new mining technologies.

Travel and Logistics:

- **Interplanetary Travel:** Traveling from Earth to the asteroid belt requires advanced spacecraft and can take several months, depending on the chosen propulsion system and the specific destination within the belt.
- **Supply Chain and Trade:** The mining colonies in the asteroid belt rely on regular shipments of equipment, supplies, and personnel from Earth and other settlements. Trade networks facilitate the exchange of resources, technology, and scientific discoveries between the Belt colonies and other factions in the solar system.
- **Infrastructure and Facilities:** Mining colonies require infrastructure for resource processing, refining, and storage. They also feature habitats for residents, research facilities, docking ports for spacecraft, and transportation systems within the asteroid belt.

The Belt is where resource wealth concentrates and where regulation is thin. Mining colonies operate far from Sol Union oversight, which makes them attractive for certain kinds of work and dangerous for others.

Outer Planets

Overview: The outer planets in the solar system, particularly the moons of Jupiter and Saturn, provide a fascinating and diverse region for exploration and scientific research in "Sol: Beyond Earth." These moons host research stations, mining colonies, and outposts that focus on various activities such as scientific exploration, resource extraction, and the study of potential habitable environments.

Research Stations: Research stations are established on the moons of Jupiter and Saturn to study their unique geological features, atmospheric conditions, and potential for life. These stations are equipped with advanced laboratories, observatories, and living quarters for the research teams. Scientists and researchers from various factions come together to uncover the mysteries of these distant celestial bodies.

Mining Colonies and Outposts: Mining colonies and outposts in the outer planets region extract valuable resources from the moons. These resources include minerals, gasses, and other materials that can be utilized for fuel production, construction, and scientific experiments. Mining operations require specialized equipment and skilled personnel to extract and process resources efficiently.

Scientific Exploration: The moons of Jupiter and Saturn provide a diverse range of environments for scientific exploration. From the icy surfaces of moons like Europa and Enceladus, which may have subsurface oceans, to the rugged terrains of Ganymede and Titan, each moon offers unique opportunities for discovery. Scientific missions include studying geological formations, analyzing atmospheric compositions, and searching for signs of past or present life.

Habitability Studies: Some research stations in the outer planets region focus on studying the potential for habitable environments on these distant celestial bodies. Scientists analyze factors such as temperature, atmospheric conditions, and the presence of organic compounds to assess the possibility of supporting life or future colonization efforts. These studies contribute to humanity's understanding of habitability beyond Earth.

Challenges and Hazards: Exploring the outer planets and their moons presents several challenges and hazards that characters must face, including:

- **Extreme Environments:** The moons of Jupiter and Saturn have diverse and extreme environments, including extreme cold, intense radiation, and low gravity. Characters must be equipped with specialized suits, protective gear, and technology to withstand these conditions.
- **Navigational Difficulties:** Navigating the vast distances between the moons and avoiding orbital hazards, such as gravitational forces and asteroid swarms, require careful planning, precise calculations, and skilled piloting.
- **Resource Management:** Resources in the outer planets region may be scarcer compared to other areas. Characters must manage their resources effectively, including fuel, life support systems, and consumables, to ensure their survival and mission success.

Points of Interest:

- **Key Moons:** Highlight significant moons of Jupiter and Saturn, such as Europa, Ganymede, Titan, and Enceladus, with their distinctive characteristics, geological formations, and potential for scientific discoveries.
- **Research Facilities:** Describe research stations and outposts established on these moons, each with its specialized equipment, laboratories, and living quarters. These facilities serve as bases for scientific exploration and resource extraction operations.
- **Unique Features:** Identify unique features of the moons, such as geysers on Enceladus, subsurface oceans on Europa, or methane lakes on Titan. These features provide opportunities for scientific investigation and may have implications for future colonization or resource utilization efforts.

Travel and Logistics:

- **Interplanetary Travel:** Traveling to the outer planets region requires advanced interplanetary spacecraft capable of long-duration missions. The journey from Earth to the moons of Jupiter and Saturn can take several years, depending on the chosen trajectory and propulsion system.
- **Supply and Support:** Research stations and mining colonies in the outer planets region rely on regular shipments of supplies, equipment, and personnel from Earth and other settlements. Supply chains and logistics networks are established to ensure the smooth operation of these remote facilities.
- **Infrastructure and Facilities:** Facilities on the moons include living quarters, research laboratories, resource processing plants, communication networks, and docking ports for spacecraft. These infrastructure elements are vital for supporting scientific research, mining operations, and human habitation.

The outer planets are the system's frontier: expensive to reach, months or years from resupply, and largely beyond the reach of Sol Union oversight.

Technology

- **Advanced Propulsion Systems:** Spacecraft utilize advanced propulsion systems such as fusion rockets, ion drives, nuclear-thermal rockets, or even exotic technologies like antimatter propulsion. These systems enable faster interplanetary travel and make exploration of distant celestial bodies feasible.
- **Artificial Intelligence and Robotics:** AI and robotics play a crucial role in space exploration and colonization. Advanced AI systems are integrated into spacecraft, research stations, and infrastructure to enhance automation, assist with maintenance and repairs, and support scientific research. Robots are used for tasks such as mining, construction, and maintenance in harsh environments.
- **Space-Based Energy Generation:** Solar power is a primary source of energy for space settlements and spacecraft. Advanced solar panels and energy storage systems harness and store solar energy, ensuring a sustainable and reliable power supply. Other forms of energy generation, such as nuclear fusion or space-based solar power satellites, are also viable.
- **Advanced Materials Science:** Carbon composites, advanced ceramics, and layered radiation shielding allow spacecraft and habitats to withstand the mechanical and radiation stresses of space that would destroy conventional materials.
- **Life Support Systems:** Highly efficient and compact life support systems are essential for sustaining human life in space habitats and during long-duration missions. Advanced recycling and filtration technologies enable closed-loop systems for air, water, and waste management. Additionally, advancements in agriculture, such as hydroponics or vertical farming, are utilized for sustainable food production.
- **Virtual Reality and Holographic Interfaces:** VR and holographic systems are standard tools for crew training, mission simulation, remote collaboration, and scientific visualization. On long transits, they also serve as primary entertainment infrastructure.

- **Advanced Spacecraft Design:** Spacecraft feature streamlined and retro-futuristic designs inspired by the aesthetics of the 1960s and '70s space race. Sleek lines, metallic finishes, and streamlined shapes evoke a sense of nostalgia while incorporating modern advancements in aerodynamics and engineering.
- **Interplanetary Spaceflight:** Interplanetary travel is routine but not fast. Fusion-powered ships cover Earth-Mars in weeks to months; outer planet transits take years. Trajectories and gravity assists are calculated by onboard computers; human crews handle the exceptions and emergencies.
- **Space Mining:** Space mining is a vital industry, driven by the need for resources to sustain the growing population and fuel technological advancements. Mining operations target asteroids, moons, and even planetary surfaces, extracting valuable minerals, metals, and volatiles. Robotic systems and automated mining equipment are used to extract, process, and transport resources back to settlements or orbital refineries. Mining colonies and outposts are established throughout the solar system, focusing on resource extraction and refining operations.

Space Habitats

Space habitats are an integral part of the colonization efforts. Inspired by Gerard O'Neill's vision, orbital habitats are large rotating structures designed to simulate gravity and provide living space for large populations. These habitats are constructed using advanced materials and feature diverse ecosystems to support agriculture and provide a comfortable living environment. Modular design principles allow for expansion and adaptation as the population grows. The habitats serve as self-sustaining communities with their own life support systems, recreational areas, and commercial zones.

- Large structures designed to house human populations in space.
- Built with advanced materials that can withstand radiation and other hazards.
- Often designed with artificial gravity systems to simulate Earth's gravity.

- Powered by various energy sources, including solar panels and fusion reactors.
- Equipped with life support systems to provide air, water, and food for the inhabitants.
- Can include recreational areas, laboratories, and manufacturing facilities.
- May be connected to other habitats or spacecraft via airlocks or docking ports.

In the Sol space there are a few types of these settlements:

- **O'Neill Cylinder**: This is a cylindrical space habitat proposed by physicist Gerard O'Neill. It consists of two large cylinders, each approximately 32 km long and 8 km in diameter, rotating around a common axis to simulate gravity. The interior of the cylinders can support a population of up to one million people.
- **Bernal Sphere**: This is a spherical space habitat proposed by physicist John Desmond Bernal. It consists of a hollow sphere with a diameter of about 1.6 km, rotating to simulate gravity. The interior can support a population of up to 10,000 people.
- **Stanford Torus**: This is a donut-shaped space habitat proposed by NASA in the 1970s. It consists of a large ring with a diameter of about 1.8 km, rotating to simulate gravity. The interior can support a population of up to 10,000 people.
- **Island Three**: This is a space habitat proposed by physicist Gerard O'Neill. It consists of a large sphere, about 1 km in diameter, with three smaller spheres attached to it. The interior can support a population of up to 10,000 people.

Strategic locations known as Lagrange Points have become home to remarkable space habitats, heralding a new era of human colonization and exploration.

- At Lagrange Points L4 and L5, approximately 60 degrees ahead and behind Luna's orbit, clusters of habitats known as "L4 cities" and "L5 cities" house permanent populations and serve as waypoints for inner-system traffic. Gravity at these points is negligible; the habitats spin for internal gravity.

- L1, positioned between Earth and Luna, hosts the “L1 city”: the primary transit hub for Earth-Luna traffic, with docking infrastructure, customs, and commercial facilities serving crews and passengers moving between the two bodies.

These habitats are supplied by asteroid resources and lunar exports, and most run close to closed-loop life support. They are the most cosmopolitan locations in the system: politically complex, commercially active, and always crowded.

Orbital Lift

Orbital lift systems (space elevators and mass drivers) are key infrastructure for moving mass between planetary surfaces and orbit. A space elevator uses a cable extending from the surface to a counterweight in high orbit; once built, it transports cargo and passengers without rocket propulsion, at a fraction of the cost. Mass drivers accelerate payloads electromagnetically along a surface-based track, launching them directly into orbit or onto transfer trajectories. Both technologies reduce dependence on chemical rockets for routine logistics.

The Space Economy

The space economy in 2090 runs on four interconnected systems. **Low-orbit factories** process raw materials from the Belt and Luna into finished components and goods, using microgravity manufacturing and solar energy that are unavailable on the surface. **Lagrange Point habitats** serve as trade hubs and research stations, positioned where spacecraft from Earth, Luna, and beyond can rendezvous efficiently. **Lunar and Belt mining operations** supply the raw materials (helium-3, water ice, metals, and rare minerals) that feed everything else. **Mass drivers** on Luna launch processed resources toward Earth orbit without expensive rocket burns.

The Sol Union governs this economy through trade agreements, resource exploitation regulations, and traffic management protocols, all of which are contested, newly written, and imperfectly enforced. Major corporations operate across faction lines; regulatory gaps are wide; and the tension between Earth-based institutions seeking control and outer settlements seeking autonomy shapes almost every commercial transaction in the system.

Mission Generator

Roll on the tables below to build a mission: combine a type, location, client, complication, hazard, reward, and contact to generate the shape of an assignment. The tables are starting points; adjust results that contradict each other or that don't fit the current fiction.

Finding Missions

Roll 1d6 to get a job offer. Spend 1₵ to re-roll.

1-2: Nothing available right now.
3-4: A mission, but high risk or low reward.
5-6: Choose between two open missions.

Mission Table

Roll d20:

1. Transport supplies
2. Repair facility
3. Explore location
4. Establish outpost
5. Extract resources
6. Investigate phenomenon
7. Capture/study specimen
8. Map terrain

9. Test technology
10. Provide medical aid
11. Rescue personnel
12. Escort VIP
13. Enforce quarantine
14. Defend from attack
15. Disable enemy base
16. Sabotage operation
17. Infiltrate facility
18. Apprehend criminal
19. Guard artifact/data
20. Destroy hazardous object

Location

Roll d20:

1. Mars base
2. Asteroid colony
3. Space station
4. Moon outpost
5. Planetside colony
6. Mining facility
7. Research lab
8. Abandoned research outpost
9. Derelict ship in asteroid belt
10. Comet

11. Jupiter/Saturn moon
12. Asteroid belt
13. Planet's atmosphere
14. Radiation zone around Jupiter
15. Solar storm
16. Deep space relay station
17. Neptune/Uranus moon
18. Kuiper belt object
19. Venus atmosphere
20. Io volcano site

Client Table

Roll d20:

1. Sol Union
2. European Federation
3. North American Block
4. African Union
5. Arabian Coalition
6. Indian Republic
7. Commonwealth of Independent States
8. People's Republic of China
9. Pacific Alliance
10. Earth Global
11. Kyushucorp
12. Pizarro Incorporated

13. Kozosoft
14. AVQ
15. Martian Colonization Initiative
16. Lunar Development Corporation
17. Venus Atmospheric Research Group
18. Belt Mining Consortium
19. Space Debris Removal Agency
20. Interplanetary Medical Association

Complication

Roll d20:

1. Vital equipment damaged on arrival
2. Unexpected radiation event at the site
3. Encrypted data: crew lacks clearance to access key files
4. Rival faction is already working the same objective
5. Solar flare disrupts electronics and communications
6. Crew member suffers acute medical crisis
7. Long-range comms blacked out for the duration
8. Life support begins to degrade
9. Evidence of deliberate sabotage found aboard ship
10. Corporate overseer orders mission abort; crew disagrees
11. Navigation data was wrong; you're at the wrong location
12. Psychological break: isolation and stress peak at the worst moment
13. Quarantine protocol triggered by unknown contamination
14. VIP observer actively interferes with operations

15. Faction politics among crew create dangerous friction
16. Mission orders turn out to be illegal under Sol Union law
17. The subject of the mission is not what the briefing described
18. A second crew is already here running the same job
19. Corporate black site discovered: client has a hidden agenda
20. The distress signal that led to this mission was a deliberate trap

Hazards

Roll d20:

1. Malfunctioning life support component
2. Uncharted debris field in transit path
3. Extreme local environment (temperature, pressure, or radiation beyond suit tolerances)
4. Experimental equipment behaves unpredictably
5. Classified hardware: illegal to possess, dangerous to ignore
6. Rival crew or faction ship on an intercept course
7. Intense solar activity flooding sensors and communications
8. Armed individuals have taken or are targeting the ship
9. Crew member becomes irrational under stress or substance use
10. VIP or client is actively undermining the mission
11. Orders received mid-mission require something unethical
12. Quarantine breach: contamination risk of unknown origin
13. Corporate agent embedded in crew is reporting everything back
14. Civilians or refugees discovered and requiring immediate assistance
15. Radiation pocket in an otherwise charted safe zone

16. Hostile ship with unclear affiliation has the crew outgunned
17. Rogue AI controlling a facility or ship refuses crew commands
18. Unexplained broadcast signal: origin and intent unknown
19. The political situation has changed; the mission is now illegal
20. The cargo or test subject is something very different from the manifest

Mission Reward

Roll d20:

1. Money
2. Valuable data
3. Fame/glory
4. Promotion
5. Rare technology
6. Natural resources
7. Political capital
8. Advanced base
9. Exclusive access secured
10. Historic discovery
11. Allies gained
12. Rival thwarted
13. Lives saved
14. Knowledge gained
15. Settlement founded
16. Route secured
17. Sample obtained

18. Cure found
19. Weapon obtained
20. Mystery solved

Contact

Roll d20:

1. Grizzled veteran
2. Brash hotshot
3. Shady corporate agent
4. Suspicious scientist
5. Zealous fanatic
6. Desperate leader
7. Unstable genius
8. Veteran deep-space recluse
9. Quirky engineer
10. Ruthless mercenary
11. Idealistic activist
12. Paranoid hacker
13. Clueless bureaucrat
14. Devious double agent
15. Sympathetic medic
16. Unlikely ally
17. Old friend/enemy
18. Defector
19. Eccentric inventor

Inspirations

These sources of inspiration offer a wealth of knowledge, ideas, and narratives that contribute to the depth and realism of the Sol: Beyond Earth game setting. While not mandatory to enjoy the game, exploring these works can provide a deeper appreciation for the themes, technologies, and challenges involved in space colonization and exploration.

- **Allen Steele, Near Space series:** these books explore humanity's expansion into space, featuring realistic depictions of space exploration, colonization, and the challenges faced by astronauts and settlers.
- **Russ Brown & Mark Waltz, GURPS Space: Terradyne:** a roleplaying game supplement that delves into the intricacies of space colonization and settlement. It provides detailed information on designing and managing space habitats and the challenges faced by their inhabitants.
- **James S.A. Corey, The Expanse:** this series presents a gritty and realistic portrayal of humanity's colonization of the solar system. The books explore the political tensions, technological advancements, and human struggles that arise as humanity spreads across different celestial bodies.
- **Paul Elliott, Orbital 2100:** a science fiction roleplaying game that focuses on the colonization and exploitation of space. The game delves into the technical aspects of space habitats, mining operations, and interplanetary travel, providing a detailed and realistic framework for space-based adventures.
- **Makoto Yukimura, Planetes:** A manga series that follows a group of astronauts working as space debris collectors in Earth's orbit. It explores the challenges and dangers of space travel, as well as the personal and ethical dilemmas faced by the characters.

- **Bruce Sterling, Schismatrix:** a science fiction novel by Bruce Sterling set in a future where humanity has colonized the solar system. The book delves into themes of transhumanism, politics, and the consequences of radical technological advancements on society.
- **Gerard K. O'Neill, The High Frontier:** a seminal work that explores the concept of space colonization and the construction of large-scale habitats known as O'Neill Cylinders. O'Neill's visionary ideas have had a significant influence on the depiction of space settlements in science fiction.
- **G. Harry Stine, Handbook for Space Colonists:** this provides a comprehensive guide to the challenges and practicalities of establishing and maintaining space colonies. The book covers various aspects, including life support systems, resource management, and the social dynamics of living in space.
- **Richard D. Johnson & Charles Holbrow, NASA SP-413 Space Settlements - A Design Study:** published by NASA, this is a detailed study that explores the design and engineering considerations for constructing and operating space colonies. It provides insights into the technical aspects of space habitats and the potential for long-term human habitation beyond Earth.
- **Arthur C. Clarke, 2001: A Space Odyssey:** Arthur C. Clarke's iconic novel, explores the potential of human evolution, artificial intelligence, and encounters with extraterrestrial life. The story delves into the mysteries of the monoliths and the transformative journey of humanity.
- **Stanley Kubrick, 2001: A Space Odyssey (film):** Stanley Kubrick's film adaptation of Arthur C. Clarke's novel, it is a visually stunning and thought-provoking exploration of human evolution, artificial intelligence, and the mysteries of the cosmos.
- **Kim Stanley Robinson, Mars Trilogy:** the trilogy presents a detailed and scientifically grounded vision of the colonization and terraforming of Mars. The books delve into the social, political, and ecological challenges faced by the colonists and offer a comprehensive exploration of the Red Planet.

- **Andy Weir, The Martian:** this novel follows the gripping story of an astronaut stranded on Mars and his struggle for survival. The novel showcases the importance of scientific knowledge, problem-solving, and resilience in an inhospitable environment.

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